EE/CE 112/112
Electric Circuits - I

with

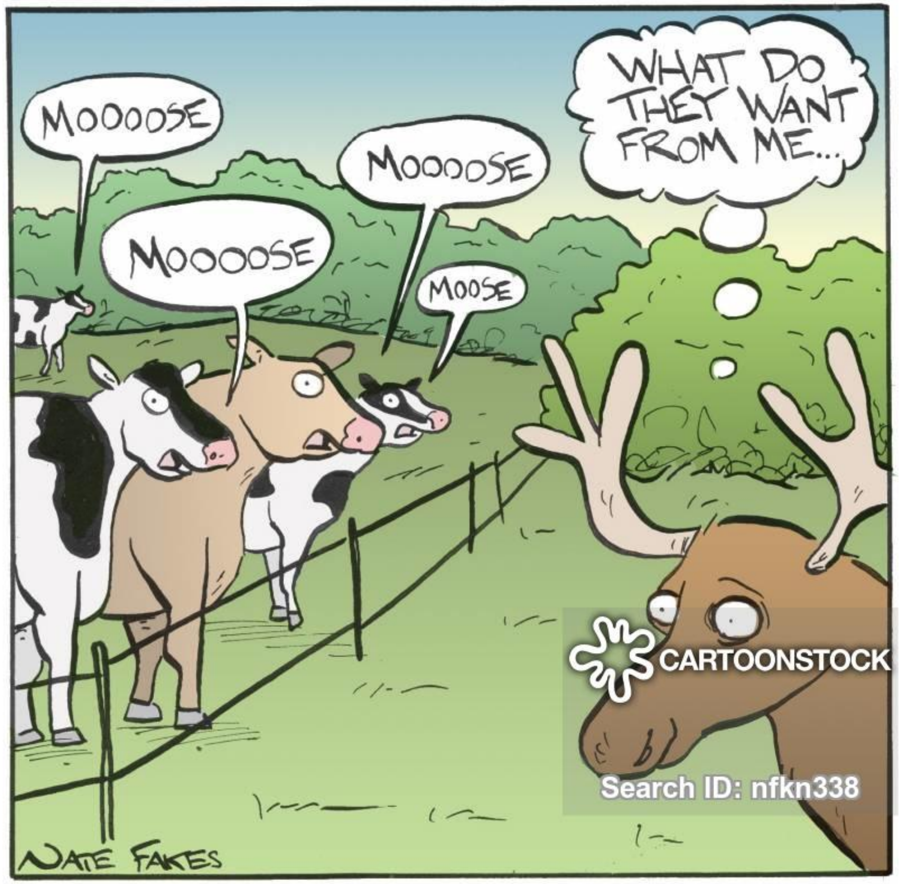
Dr. Naveed R. Butt

@

Habib University

Vocabulary

- زبانِ یارِ مَن تُرکی



What a moose hears

Some of the terms we will be looking at…

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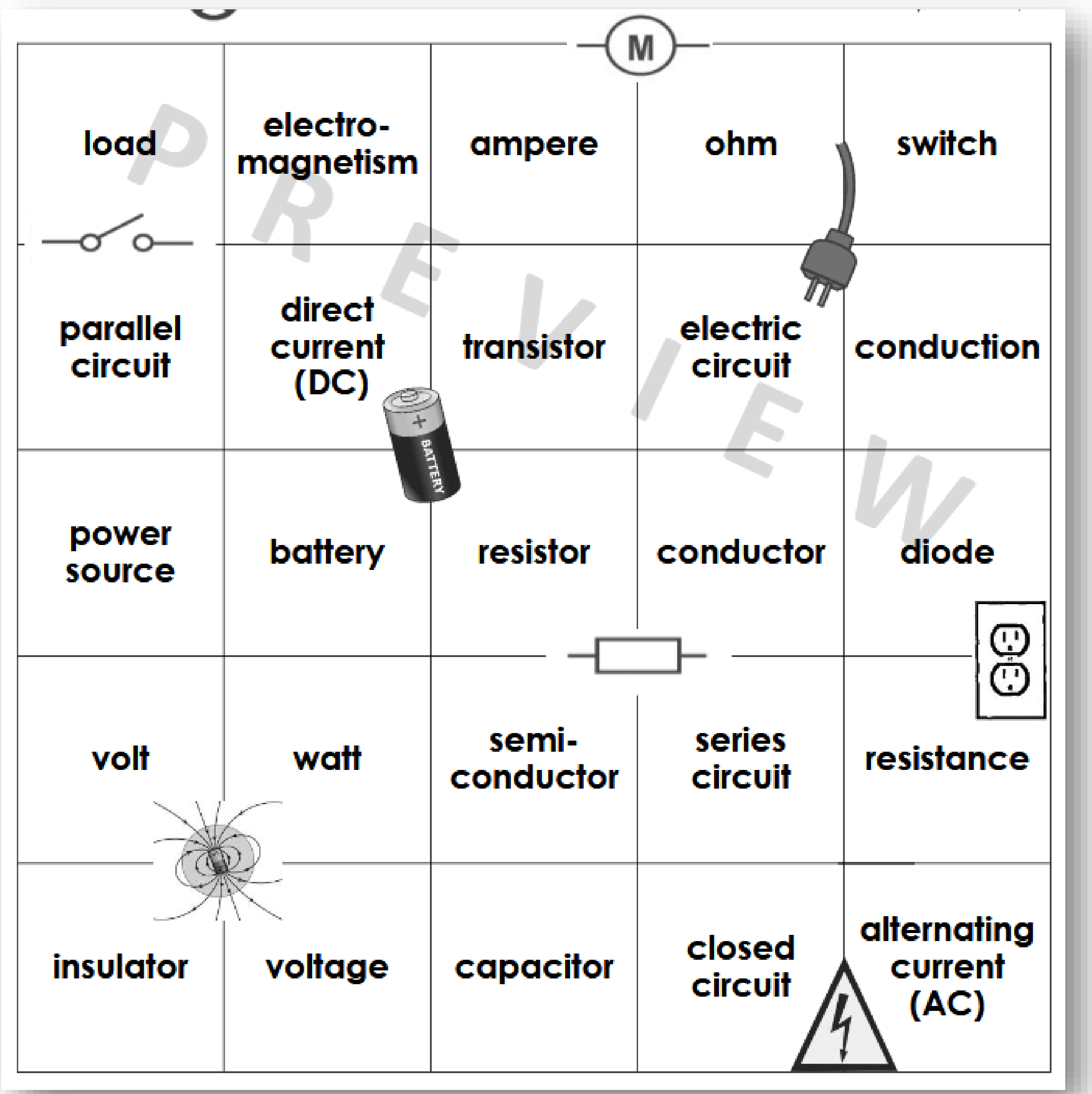

load	electro-magnetism	ampere	ohm	switch
parallel circuit	direct current (DC)	transistor	electric circuit	conduction
power source	battery	resistor	conductor	diode
volt	watt	semi-conductor	series circuit	resistance
insulator	voltage	capacitor	closed circuit	alternating current (AC)

“Electric Circuit”

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The Race Track…

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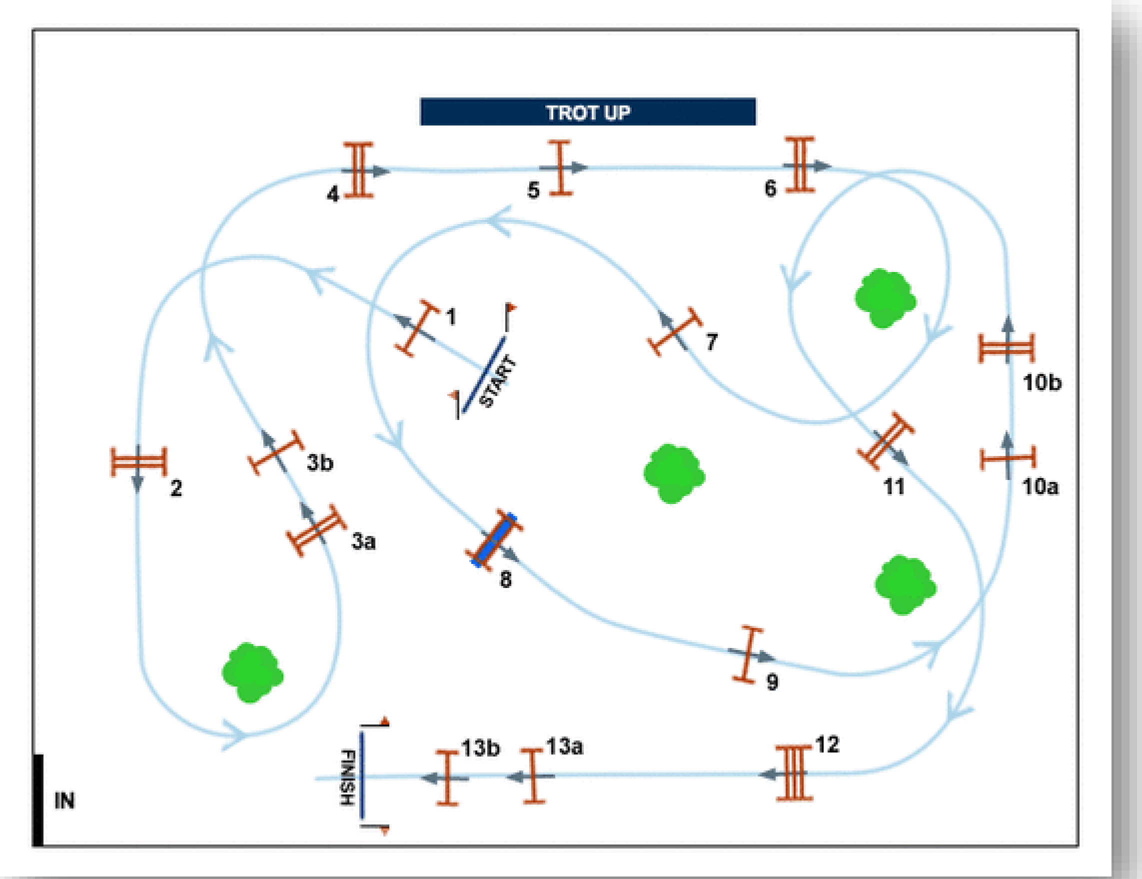
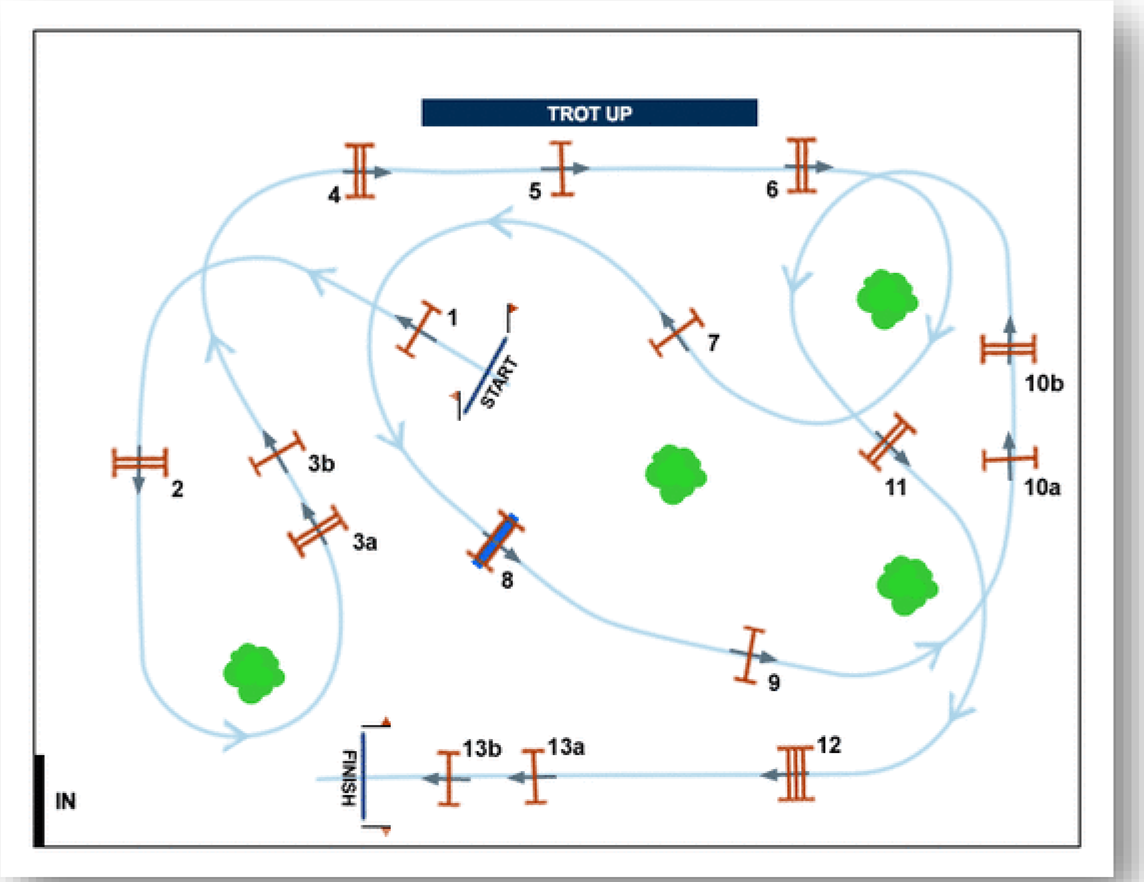


"Electric Circuit"

The Race Track…

Abstraction (simplified diagram) of the track.



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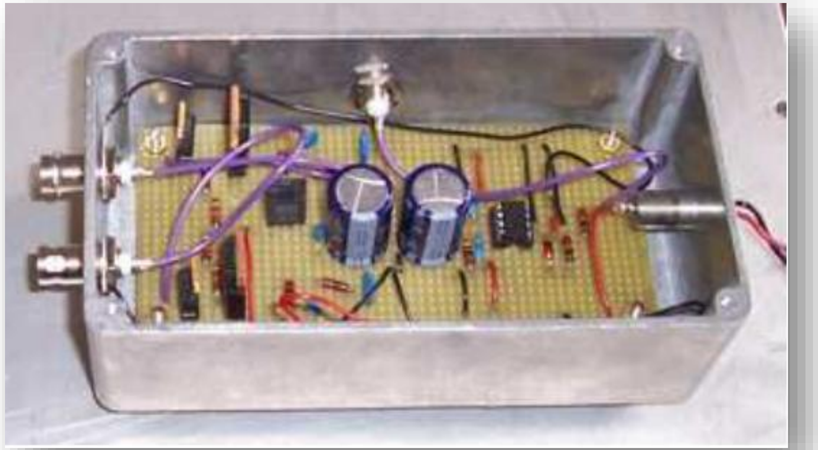

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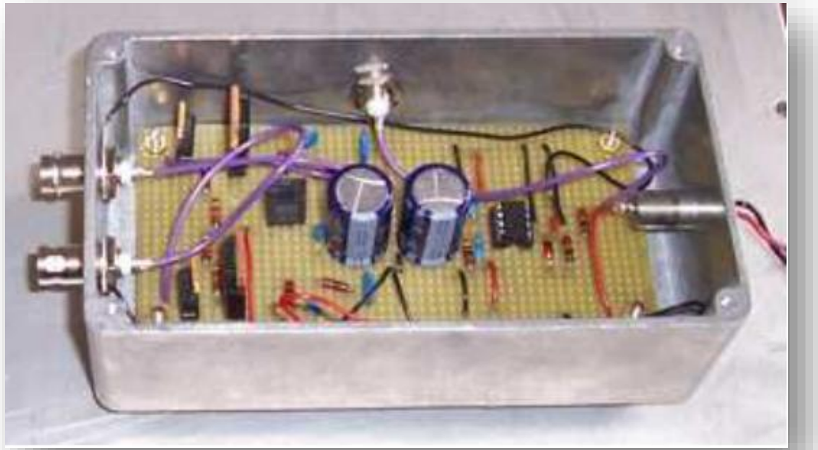

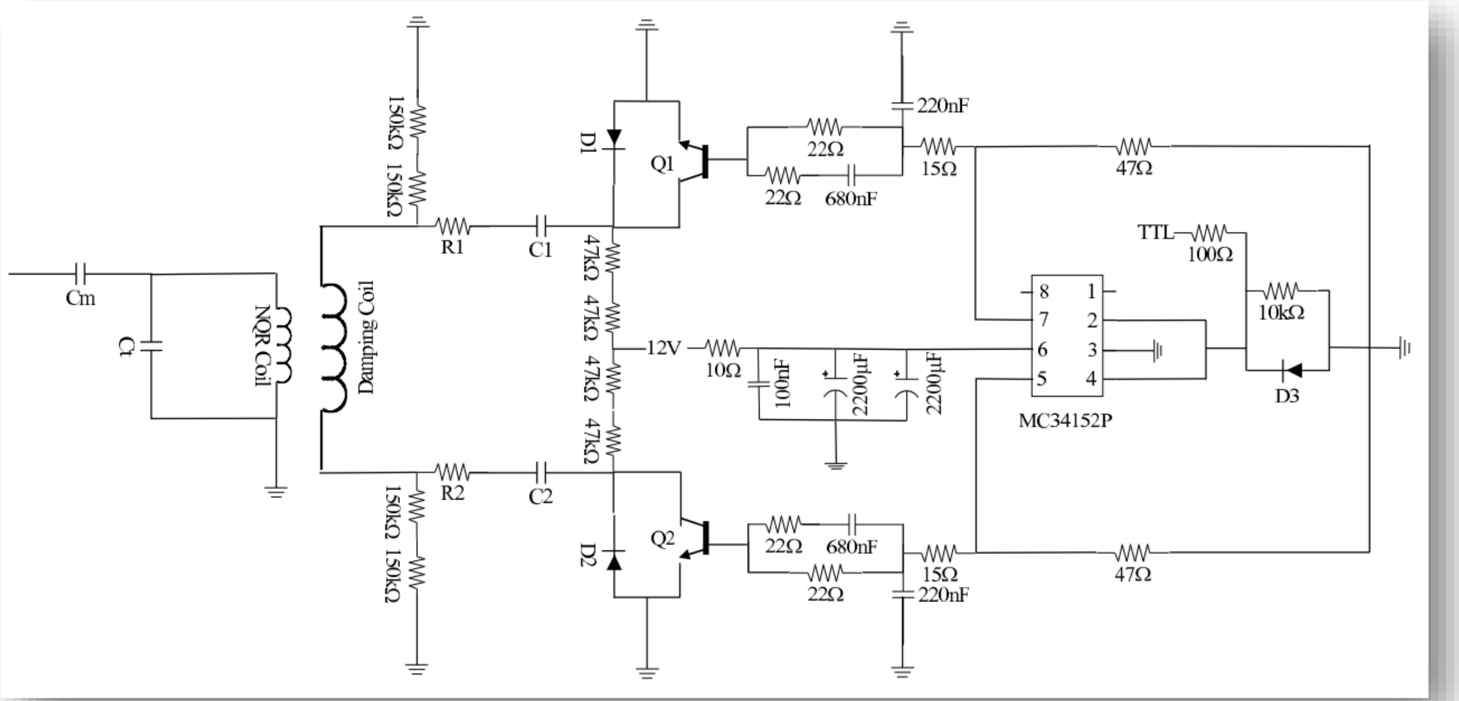

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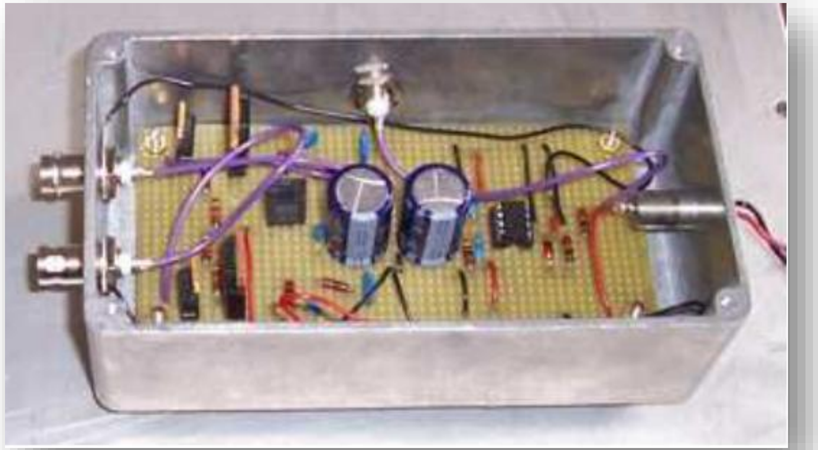

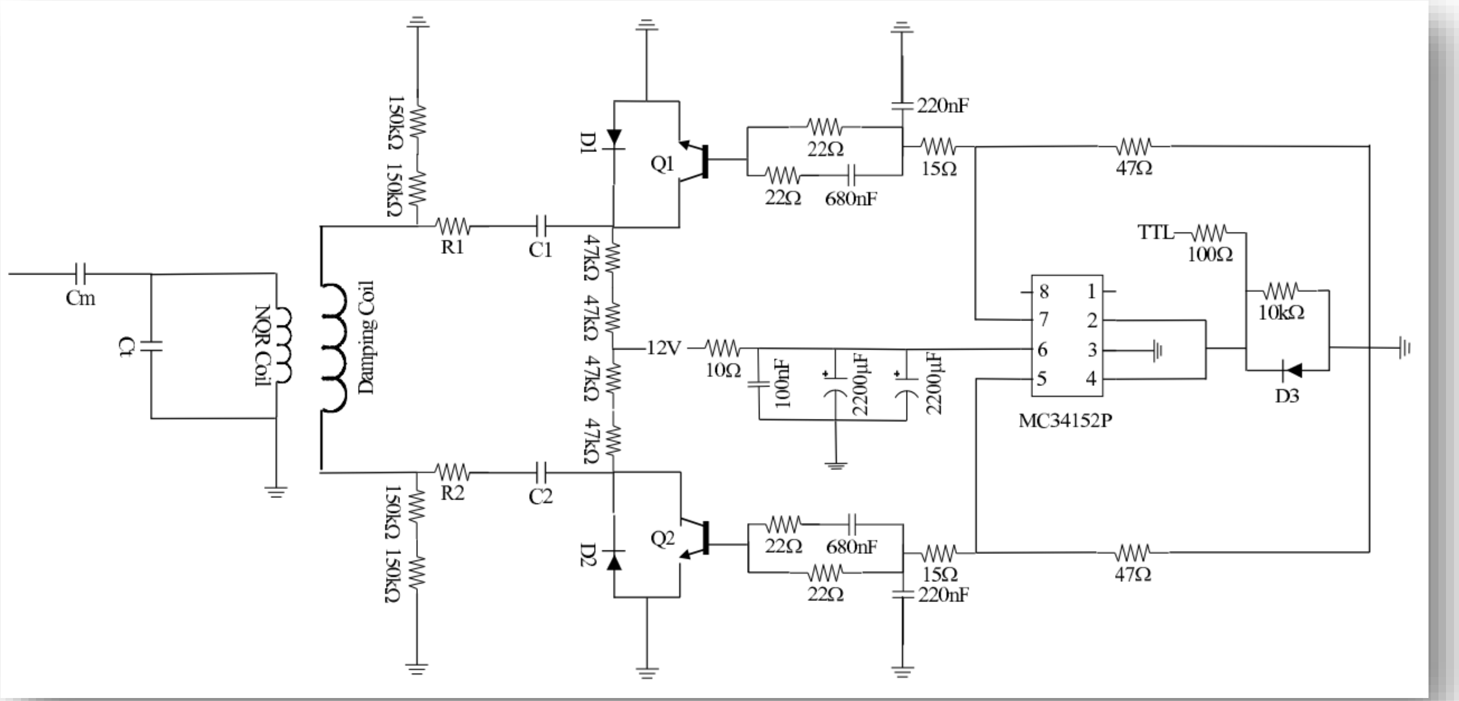

Formally speaking…

An **electric circuit** is an interconnection of electrical elements.

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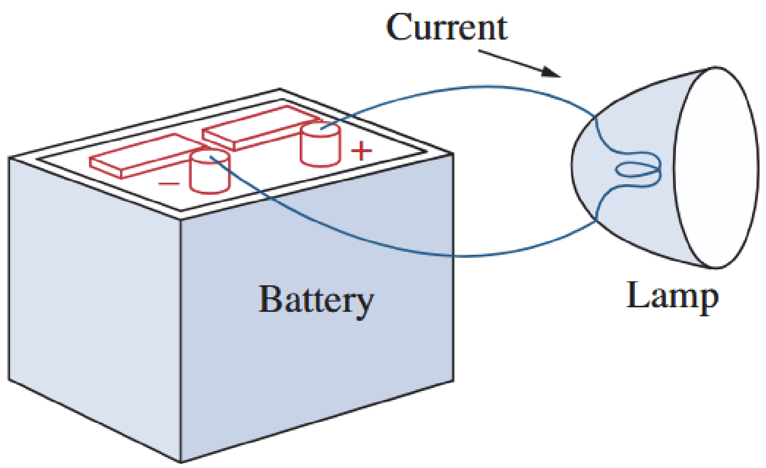

Could be as simple as this…

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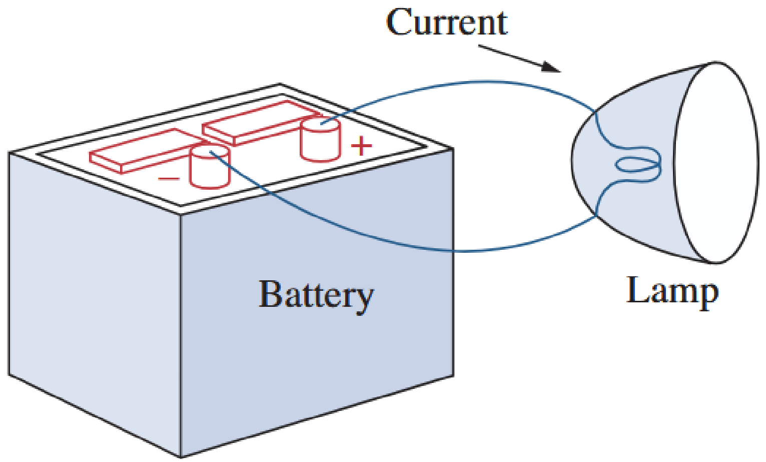

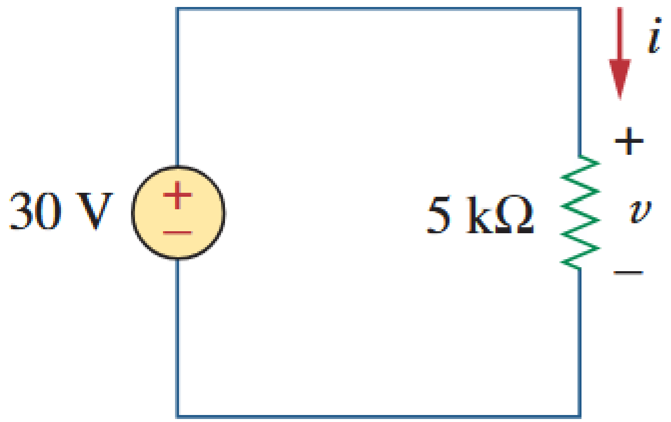

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“Charge” The Workhorse…

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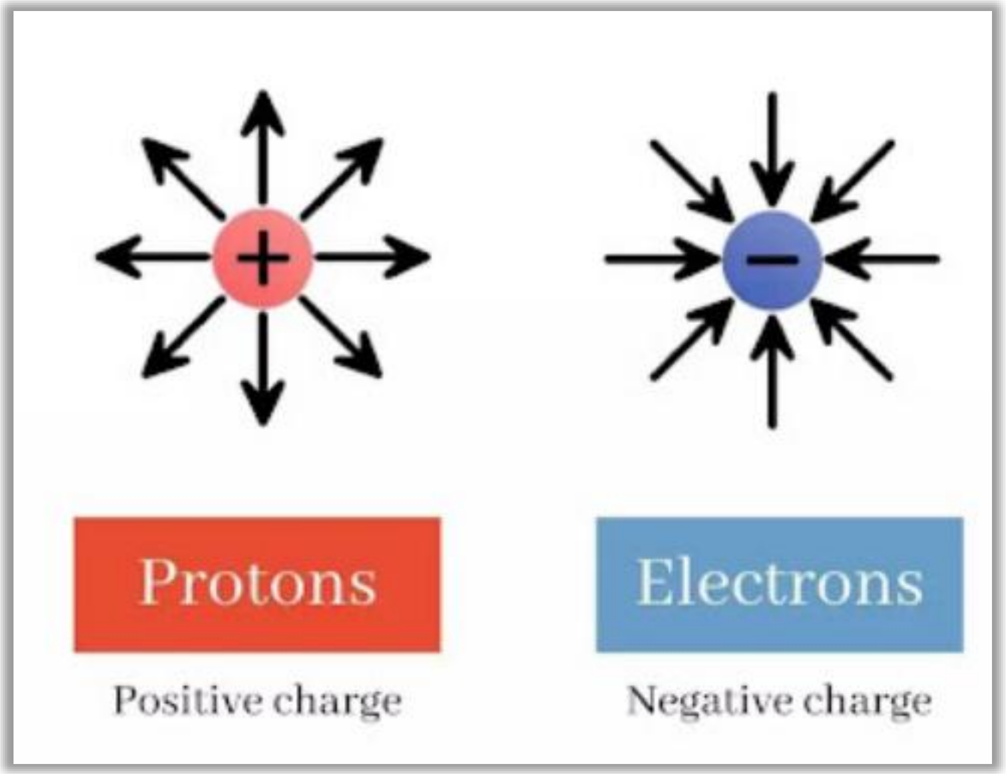
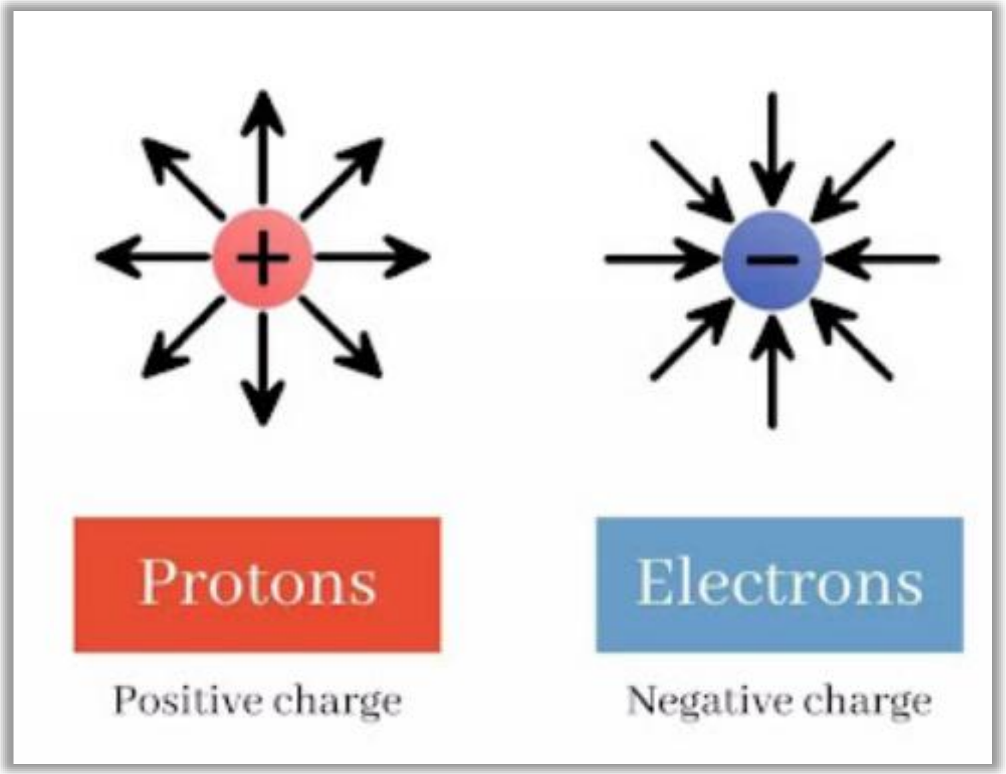


"Charge" The Workhorse…

Q. What do the arrows represent?



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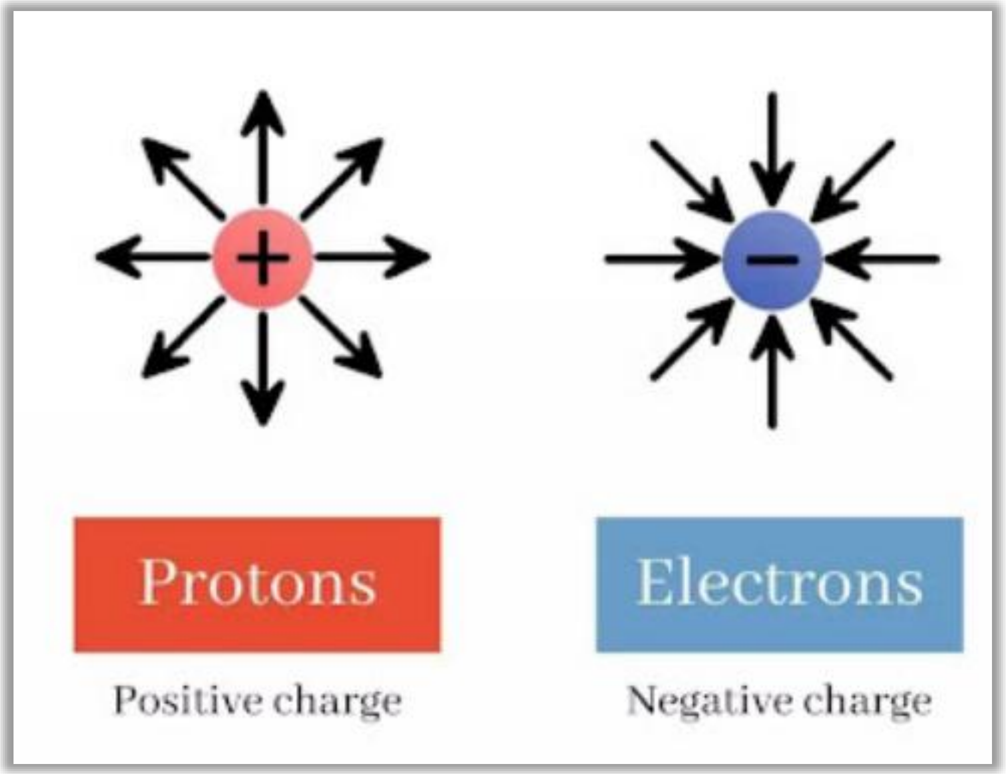

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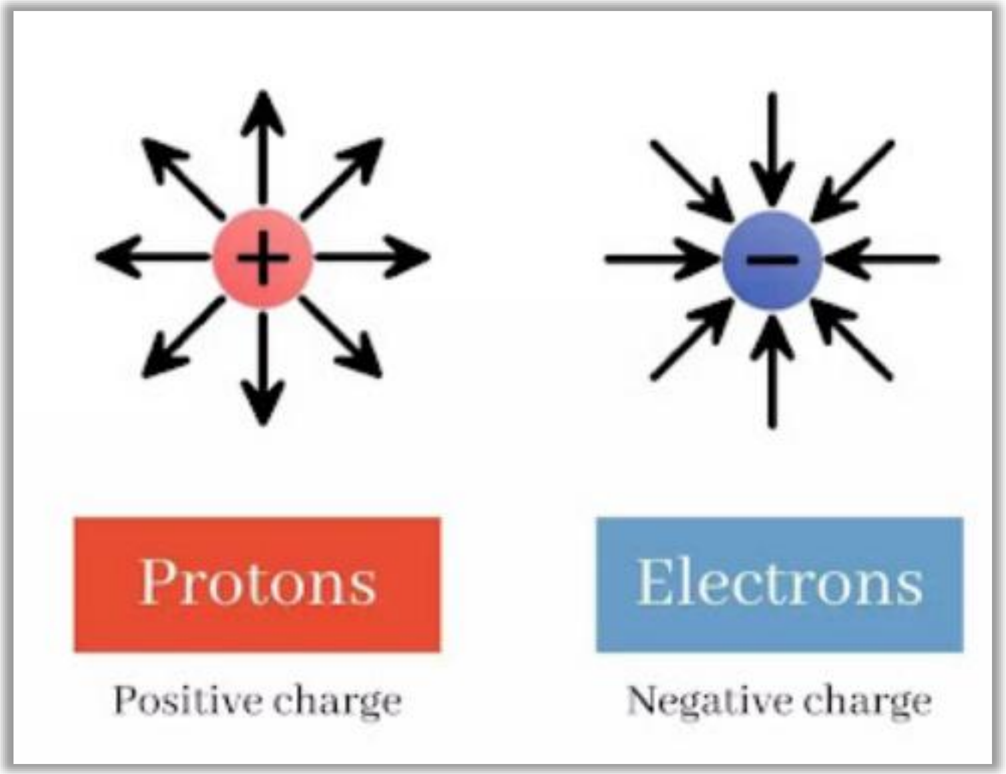

By ***convention*** we show direction of force that a positive charge would experience.

“Charge” The Workhorse…

We know from basic Physics…

The Sub-atomic Particles			
Relative size	Name	Mass (Kg)	Charge (C)
	Proton	1.67×10^{-27}	$+1.602 \times 10^{-19}$
	Neutron	1.67×10^{-27}	0
size exaggerated	Electron	9.11×10^{-31}	-1.602×10^{-19}

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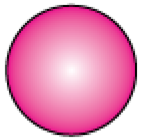
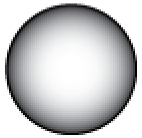
	Proton	1.67×10^{-27}	$+1.602 \times 10^{-19}$
	Neutron	1.67×10^{-27}	0
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The claim that electron is "negatively" charged is also a convention!

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Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).

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"Quantization" of charge.

“Charge”

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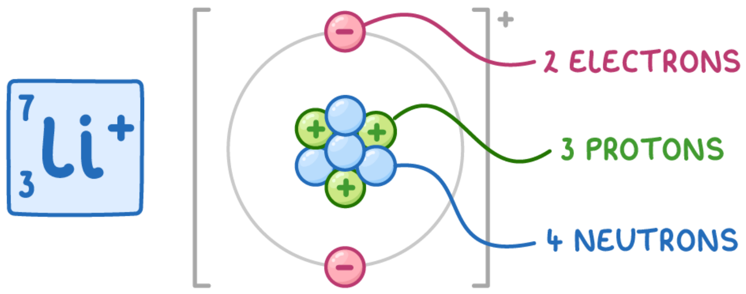

Net charge on this ion is due to one extra proton, so e.

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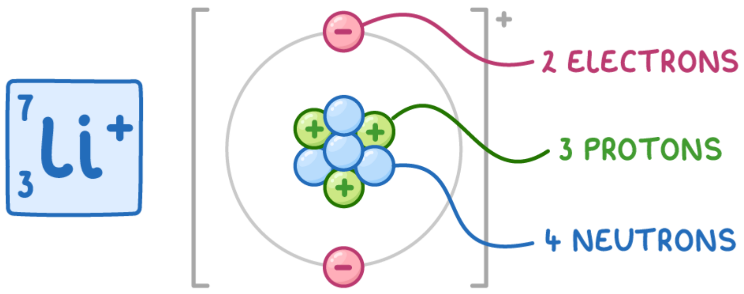

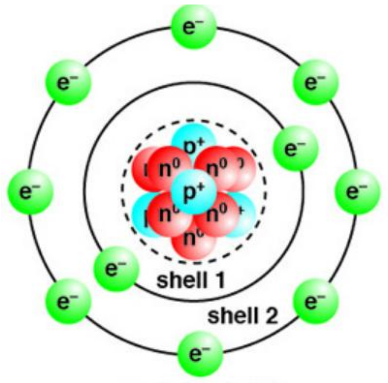

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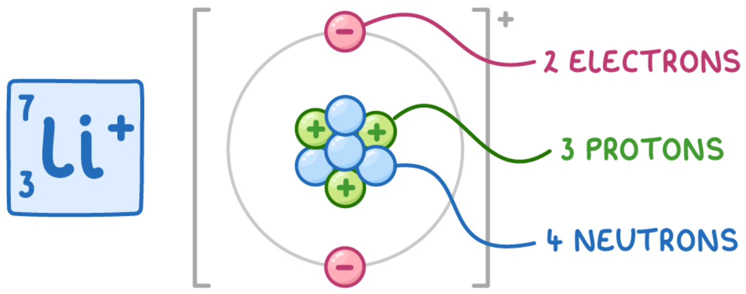

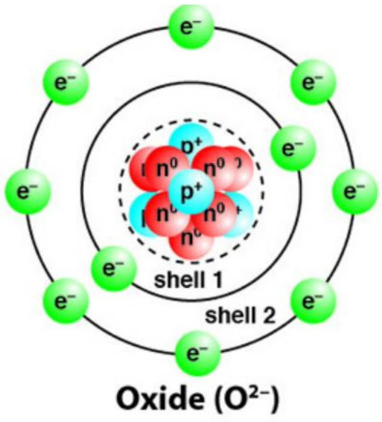

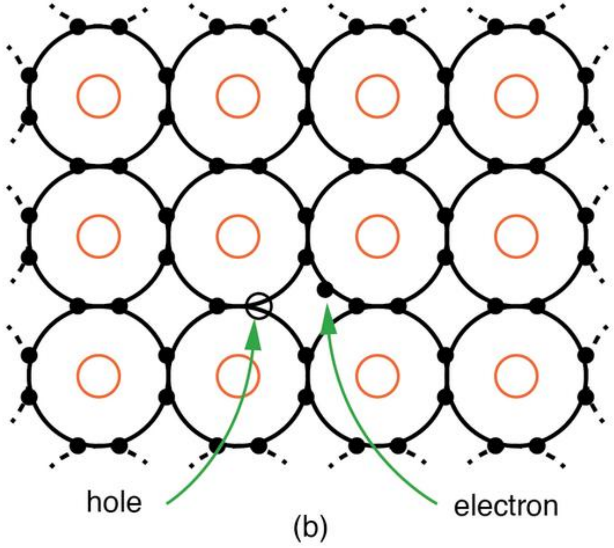

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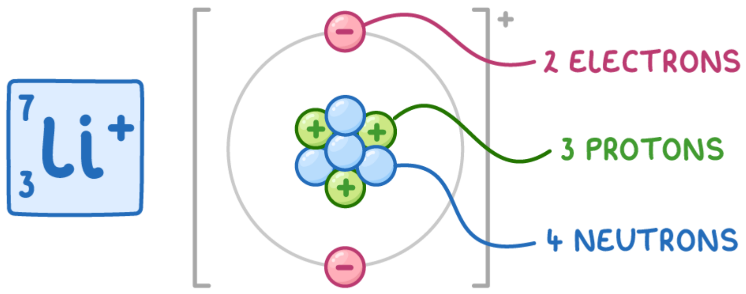

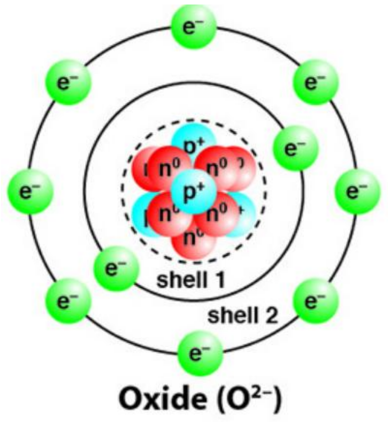

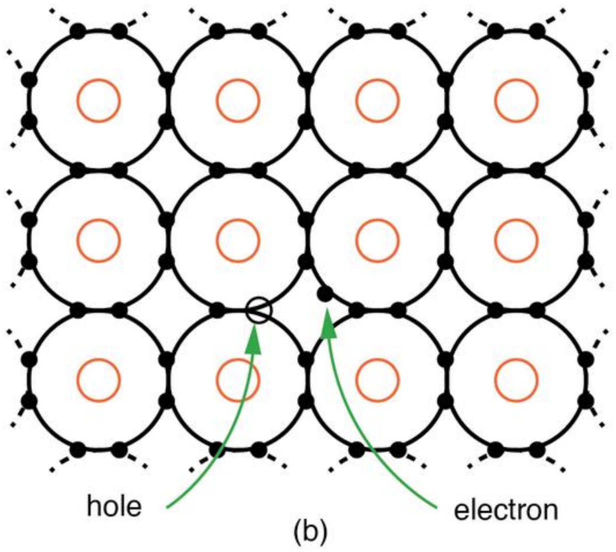

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"Hole" in a semiconductor crystal lattice is absence of an electron, so represents a charge equivalent to a proton, i.e. e.

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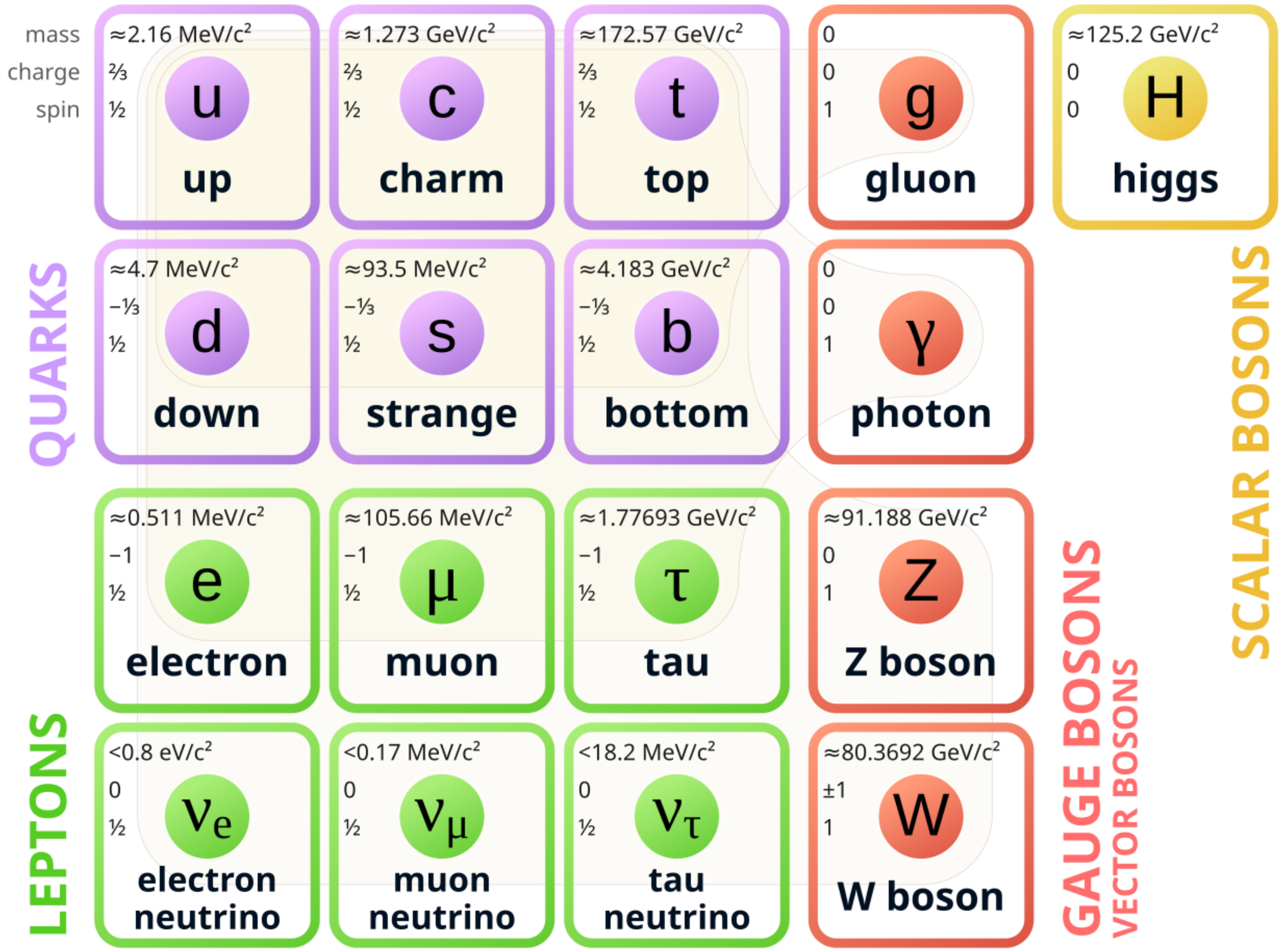

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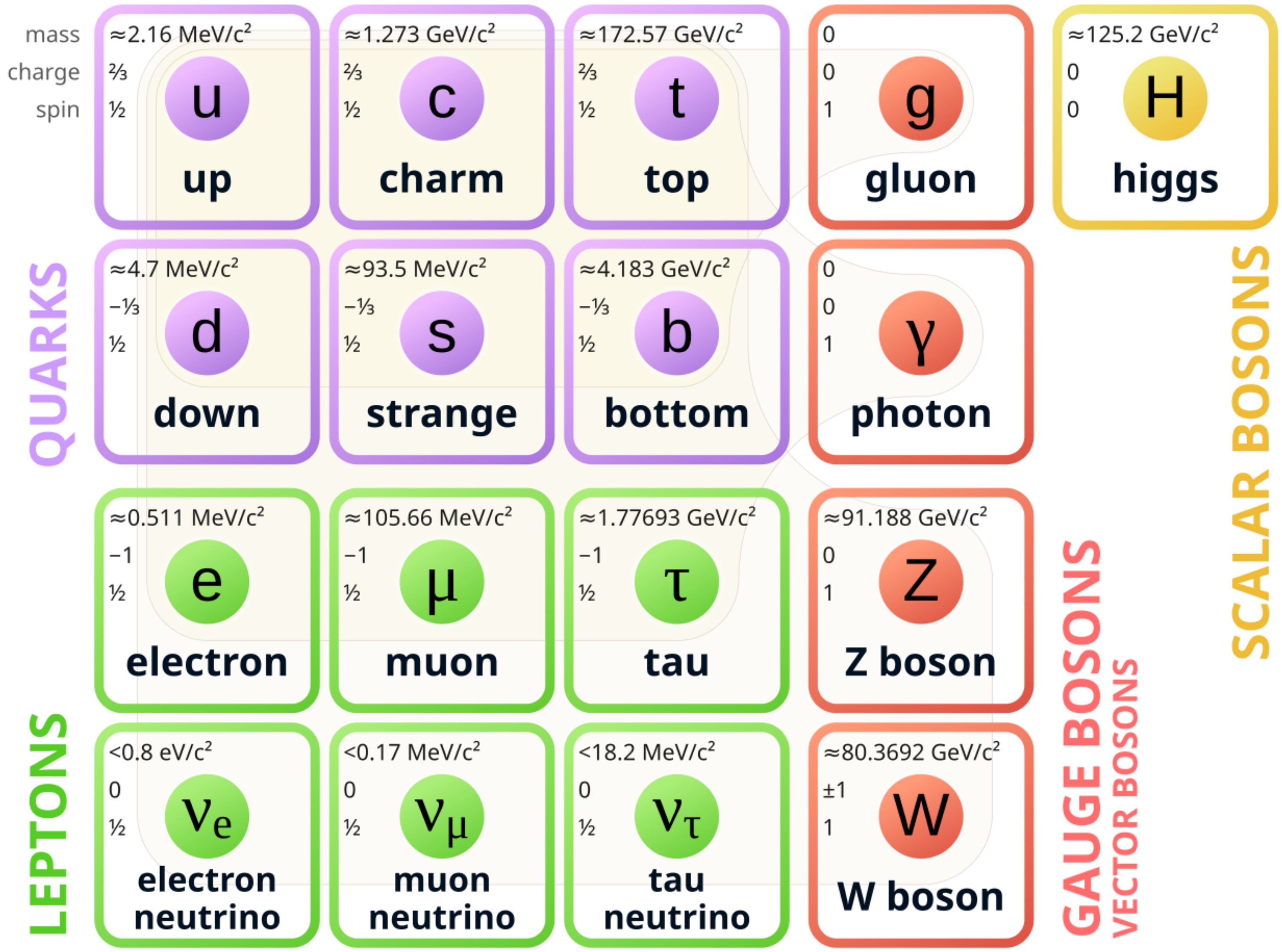
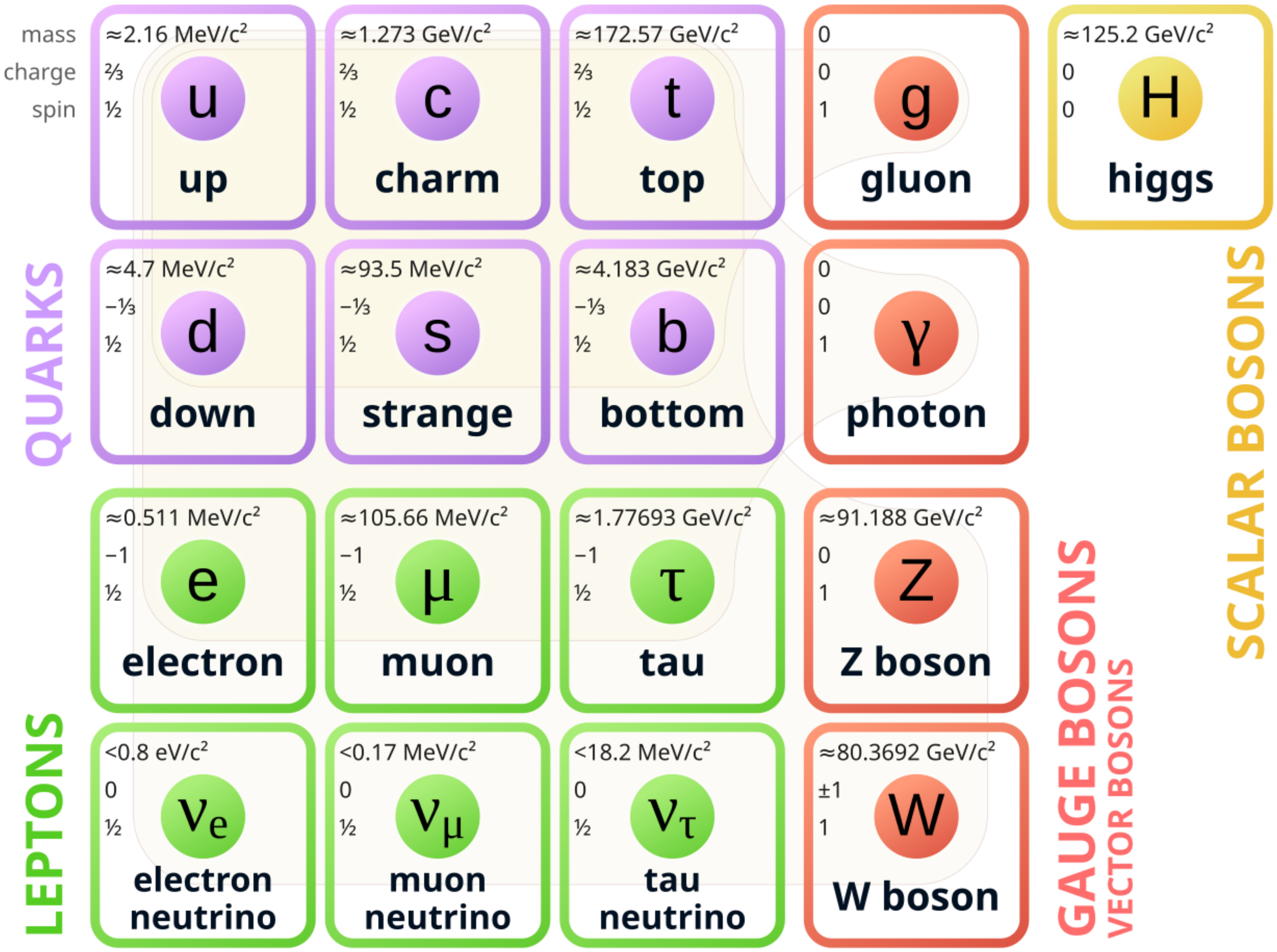
- Although quarks have fractional charges ($\frac{2}{3}e$ etc.), these are never observed in nature in isolation.



HW: What about photons, leptons, quarks etc.?

- Although quarks have fractional charges ($\frac{2}{3}e$ etc.), these are never observed in nature in isolation.

- And their grouping always leads to charges that are integer multiples of e (including 0, as in a neutron).



“Current” and **“Voltage”**

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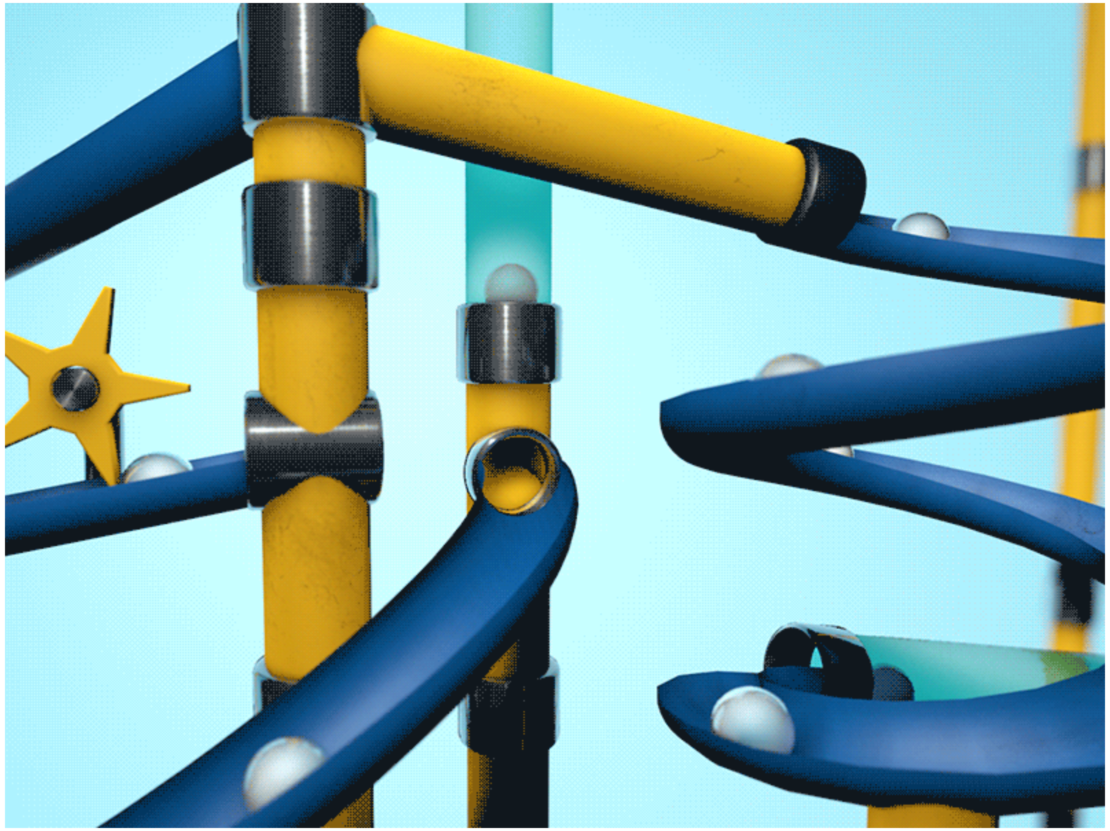

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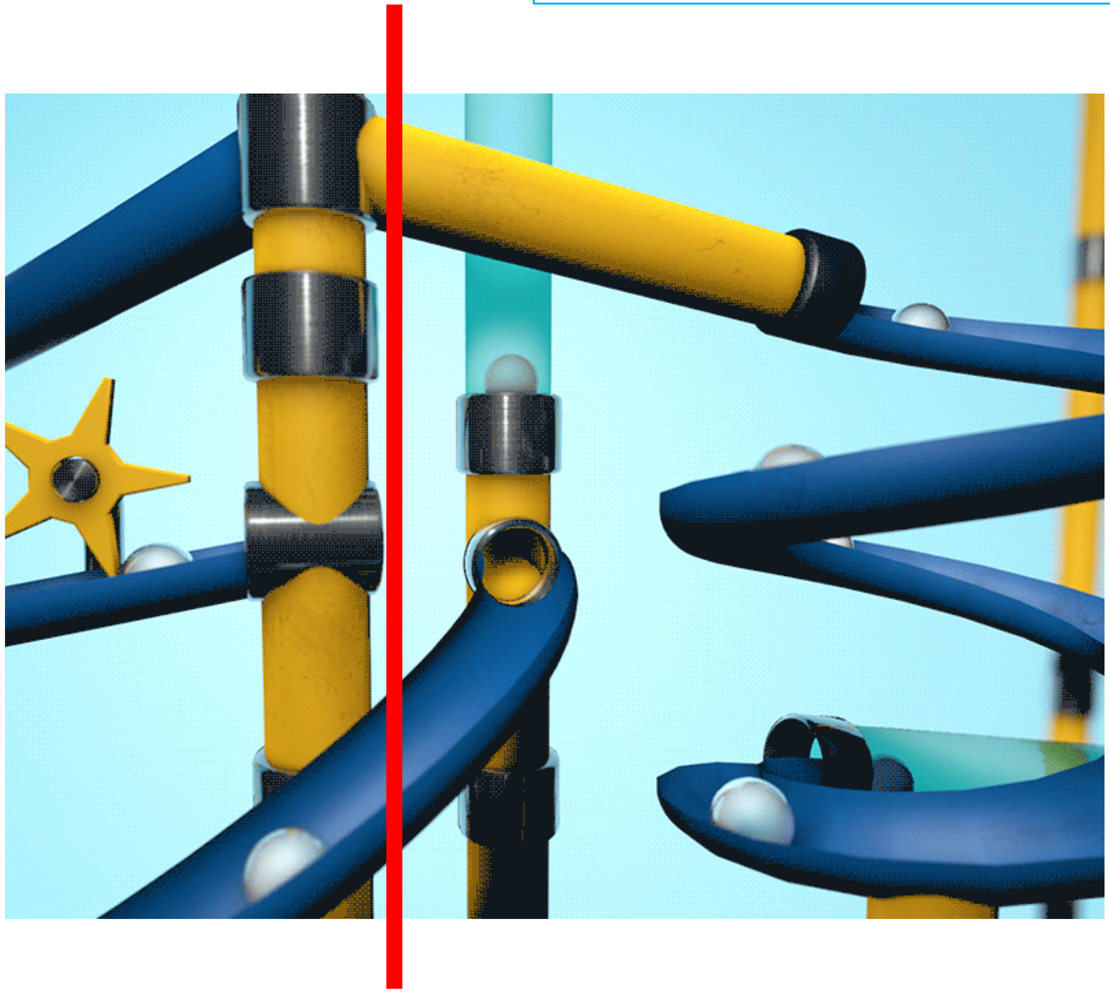

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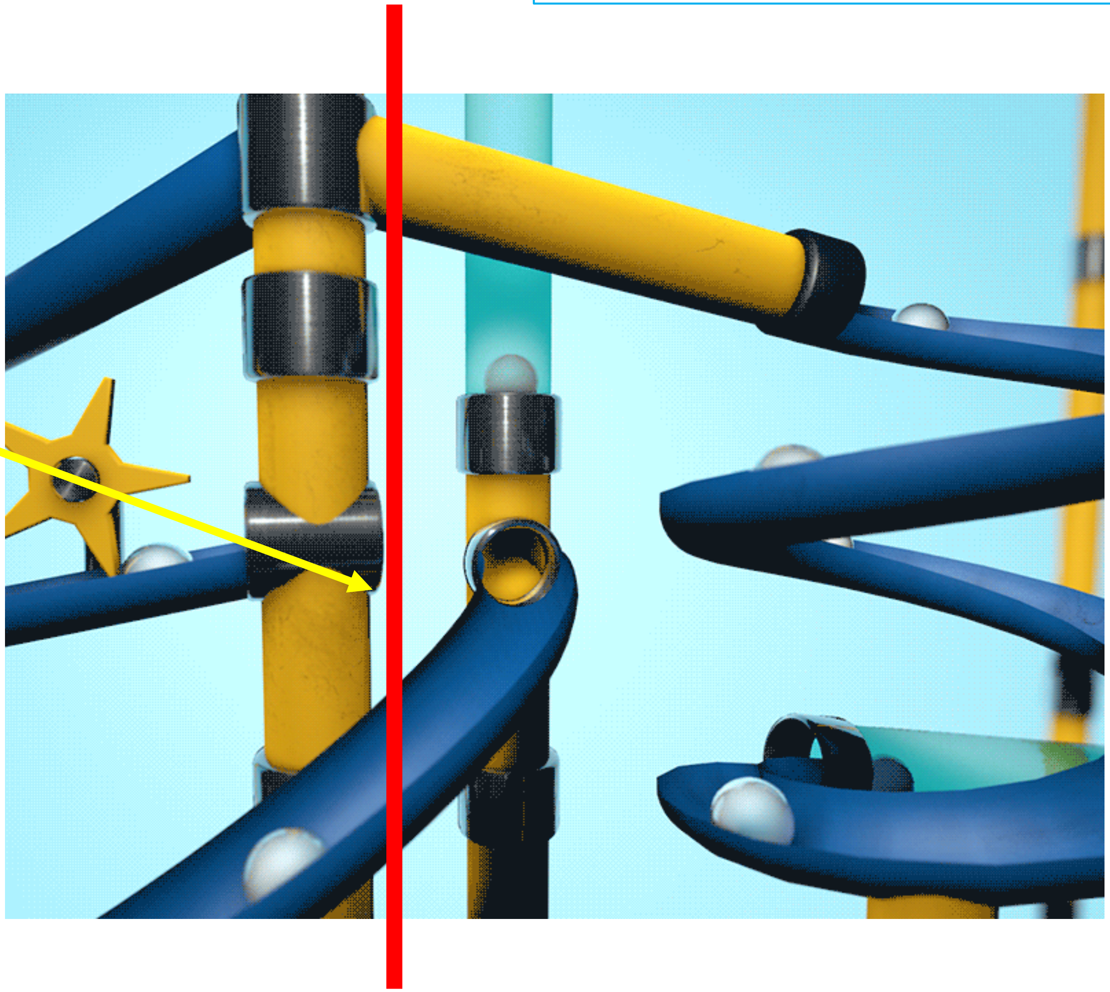
How many coulombs of net charge flowing across a certain point per second?



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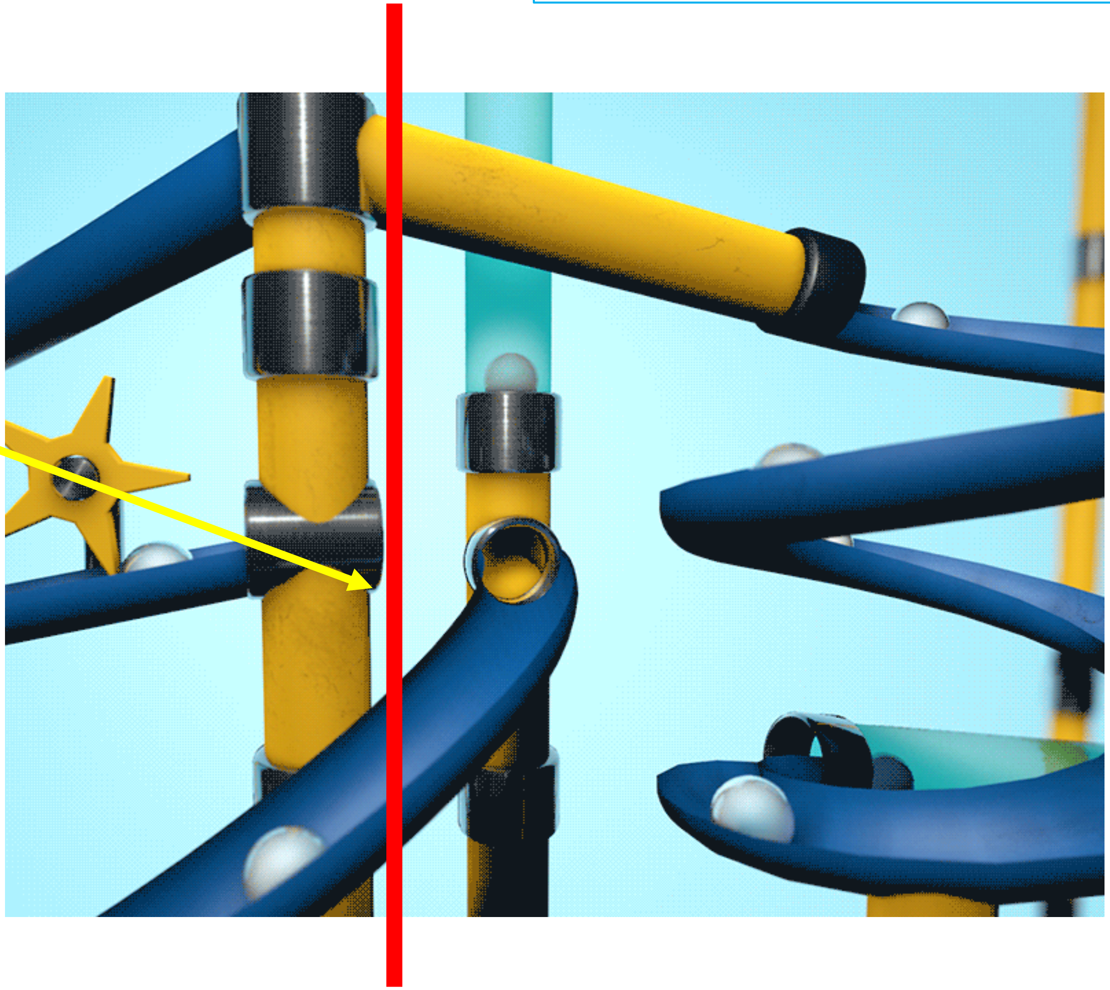

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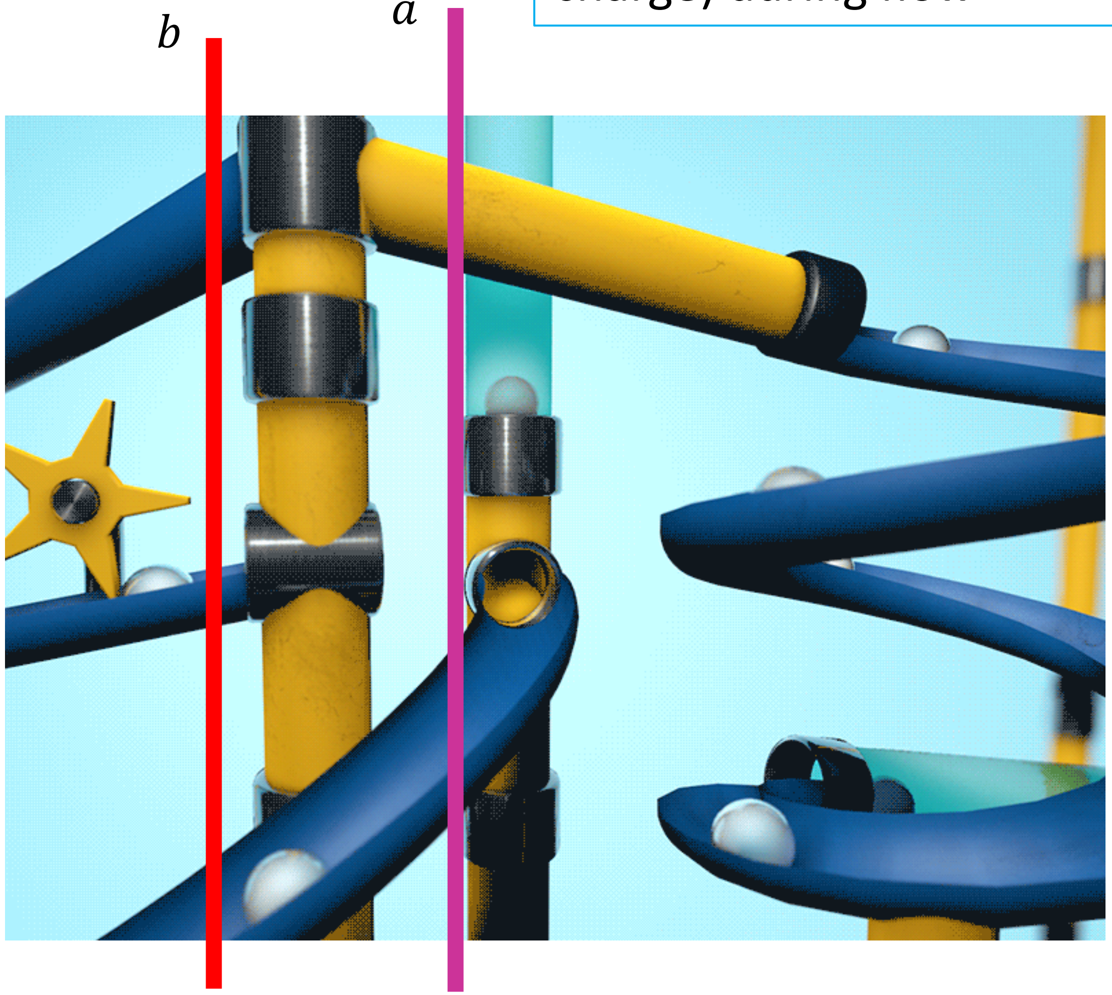

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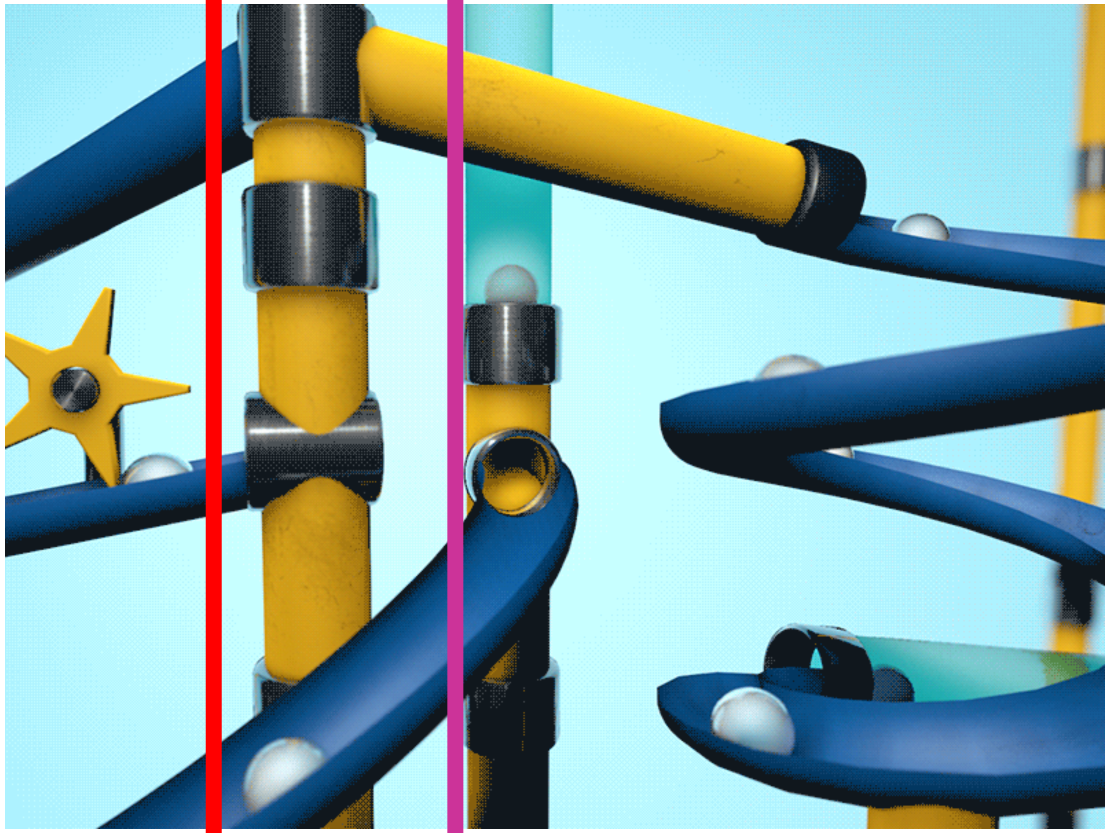
b a



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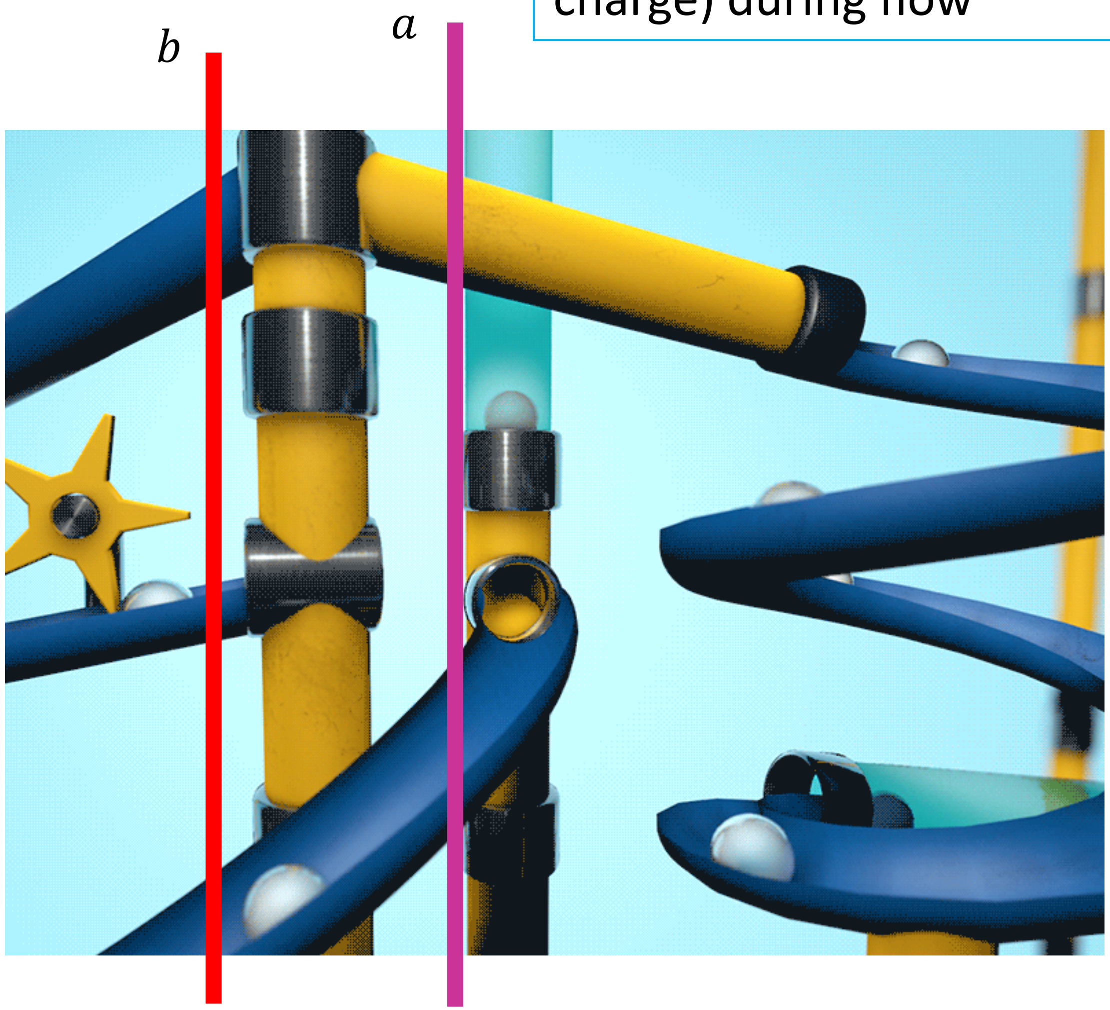

- How much energy is lost by a unit charge as it goes from a to b?

- Or, how much energy has to be provide to a unit charge to get it from b to a?

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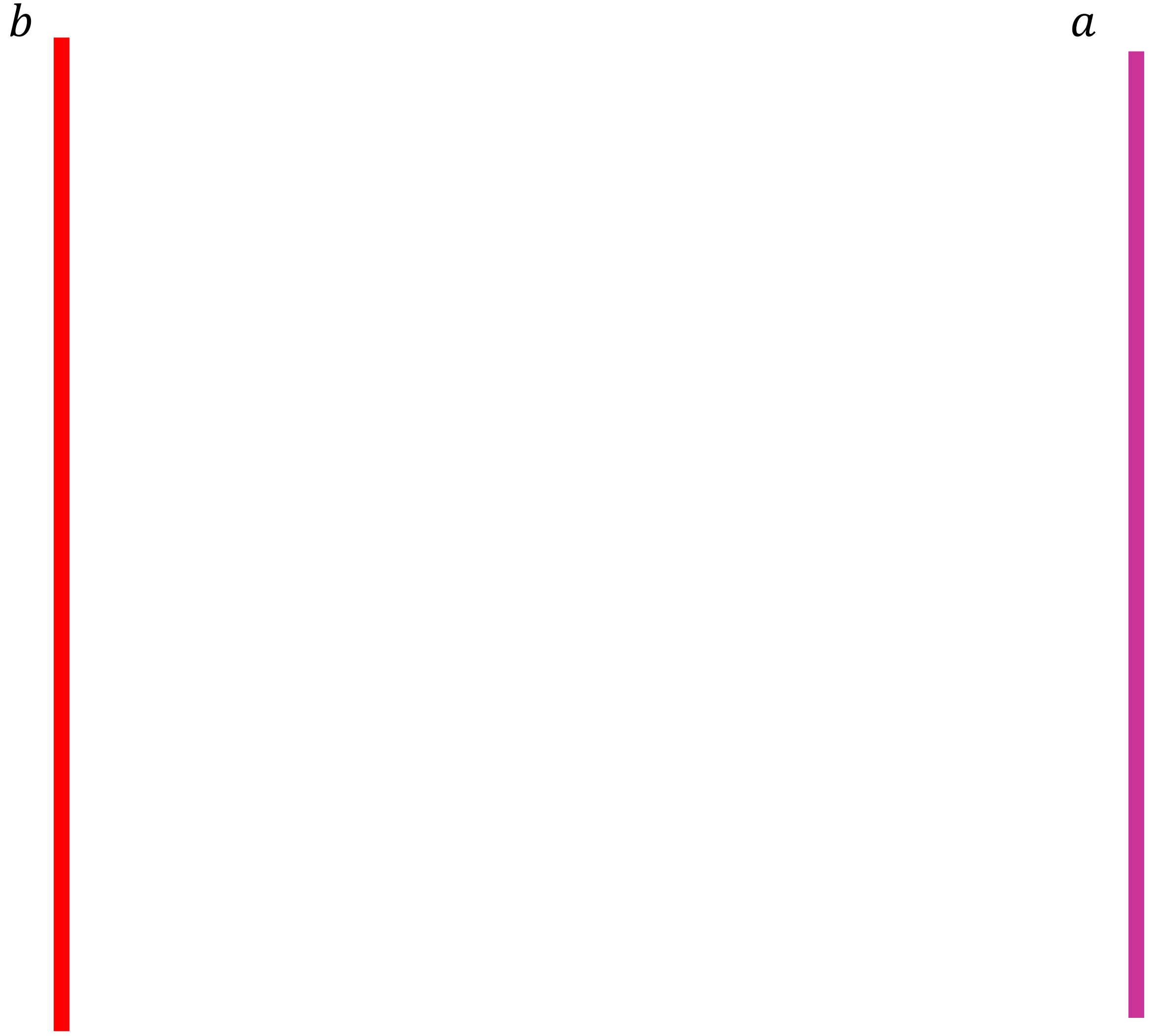
Well, without that the question "how much energy is spent in getting from a to b" would be ambiguous!



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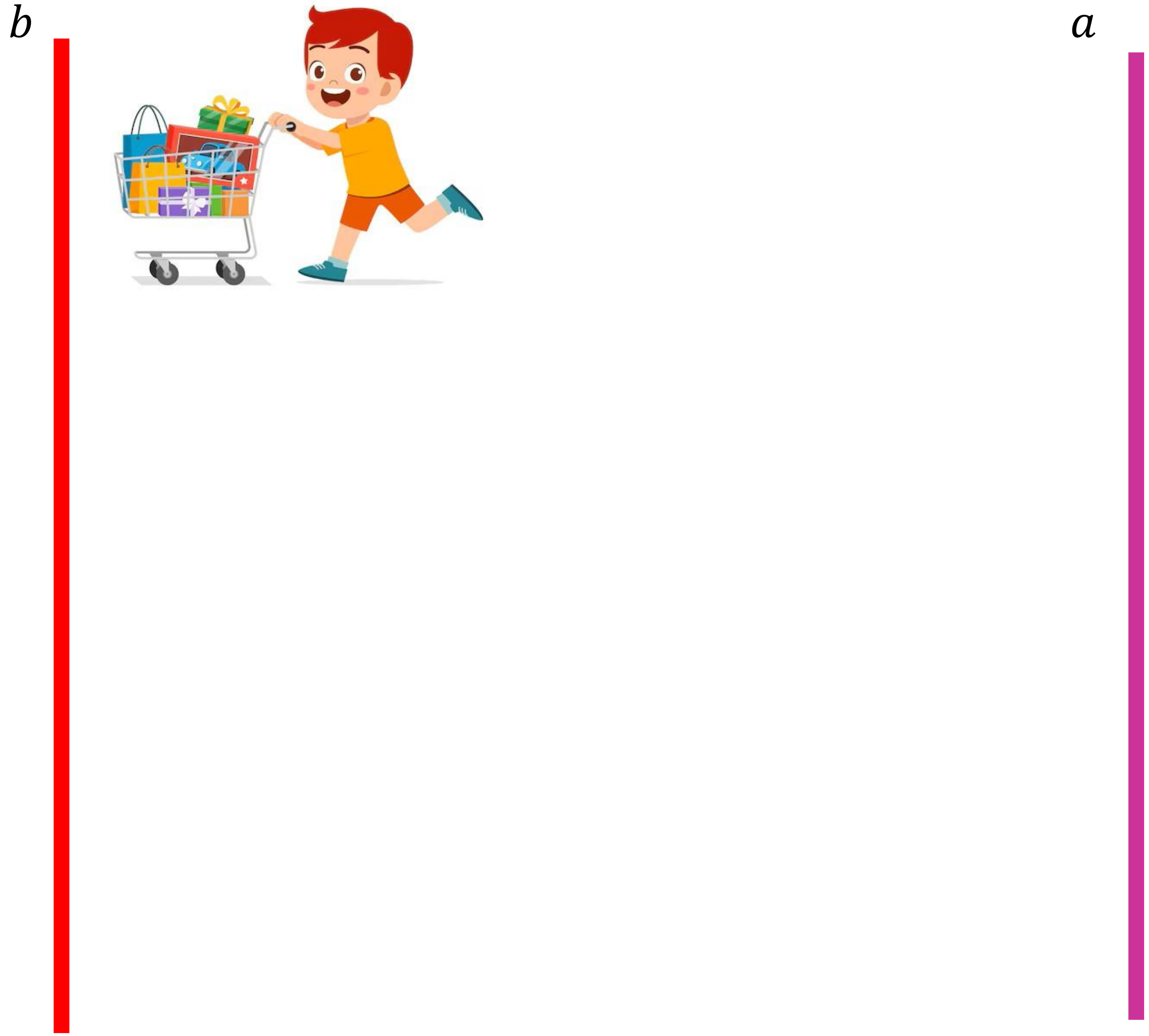

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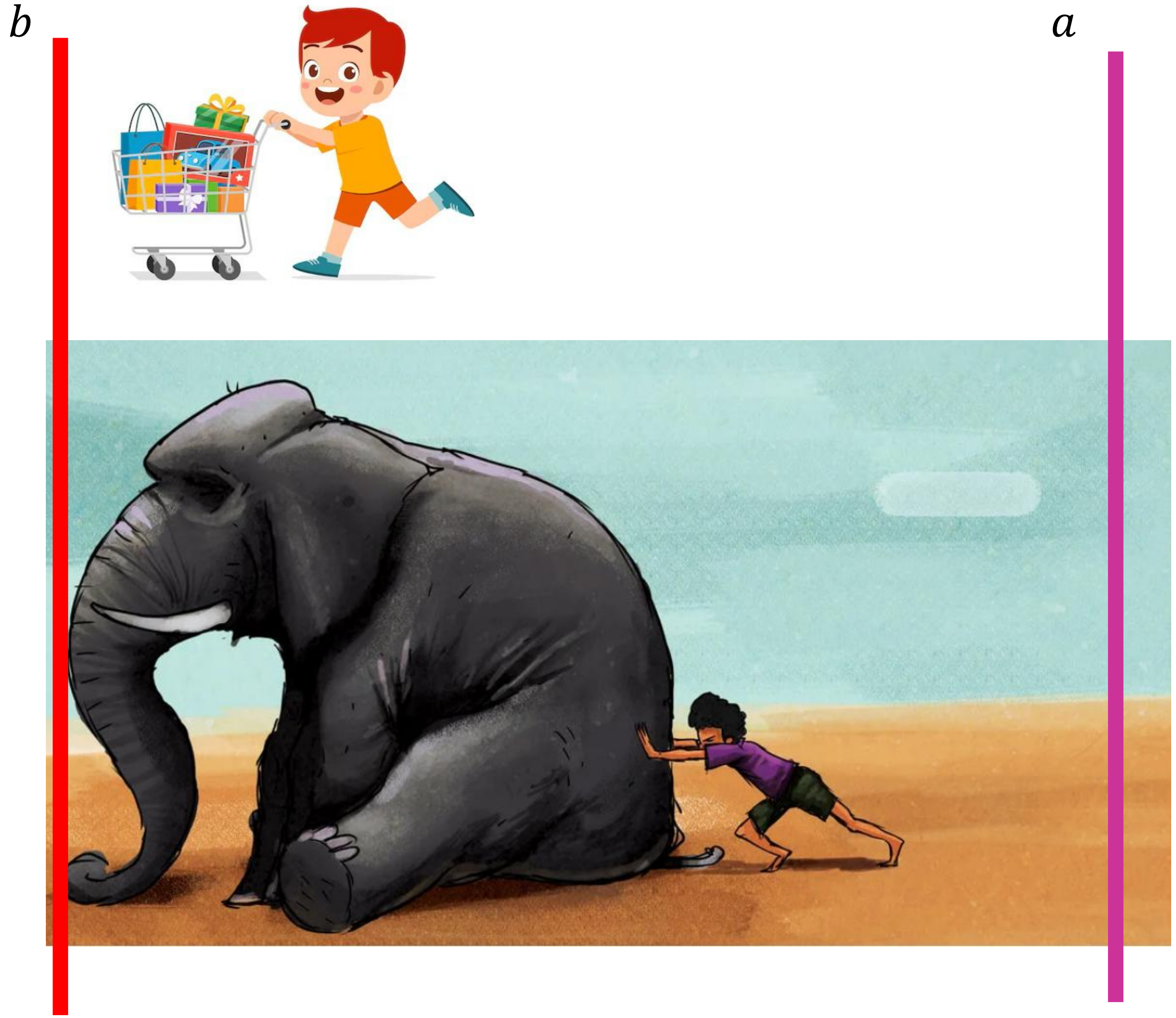
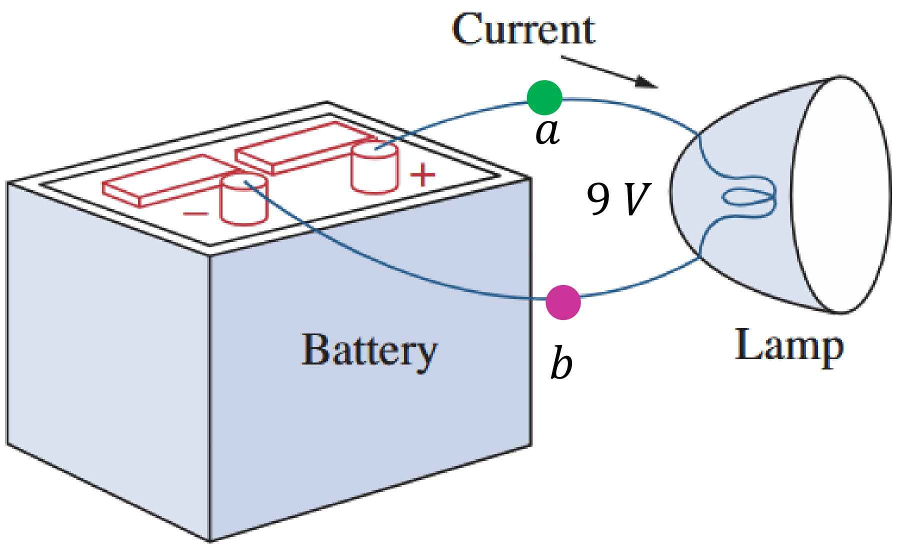


The Rise and Fall of Potential in a Circuit…



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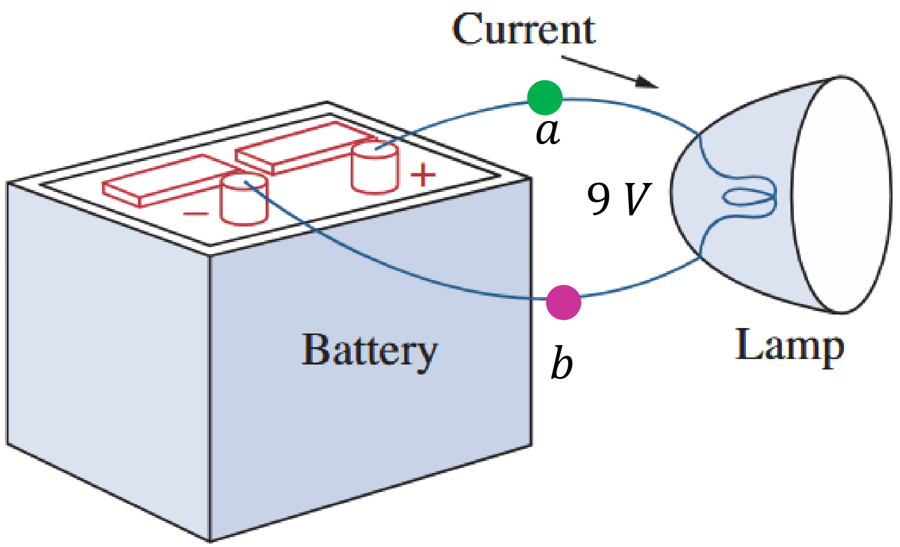

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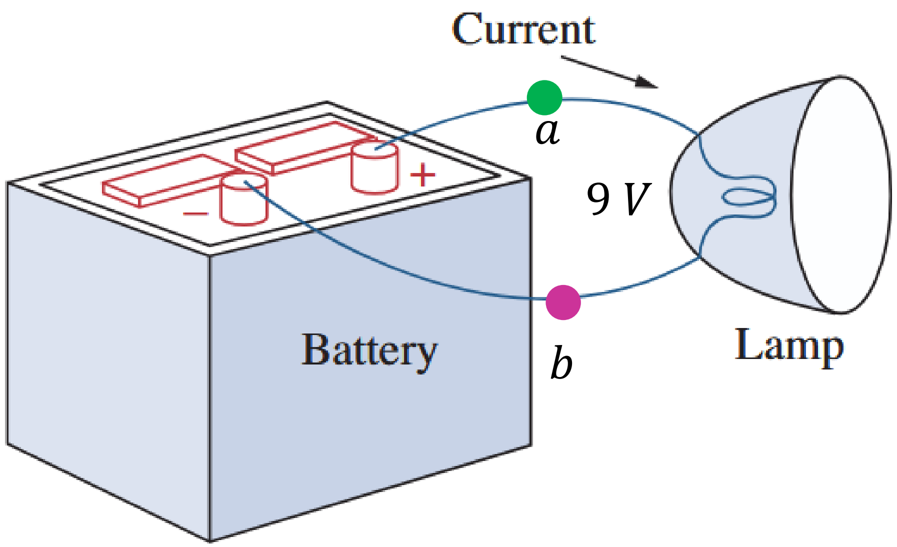

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  - As the charge flows (clockwise) from a to b, the potential drops by $9\ V$.

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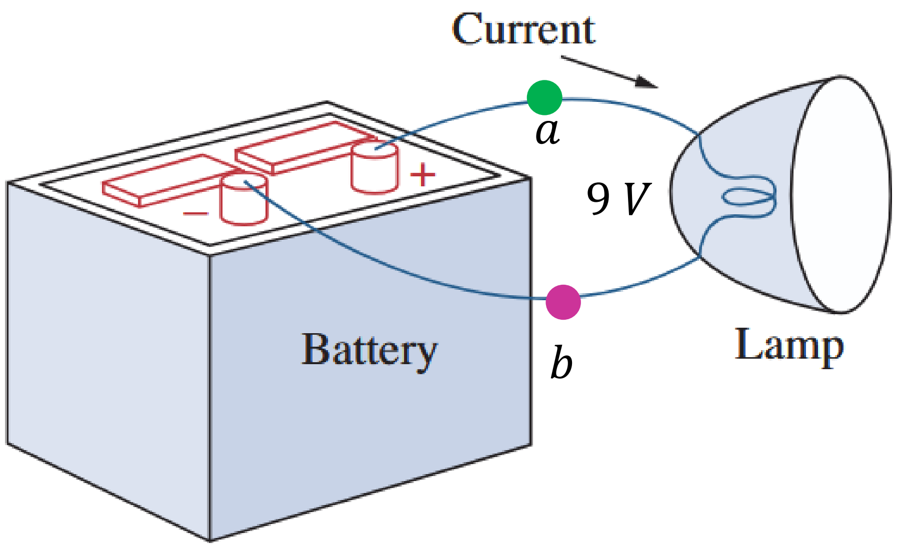

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Recall: we've already seen what happens inside the battery and the lamp!

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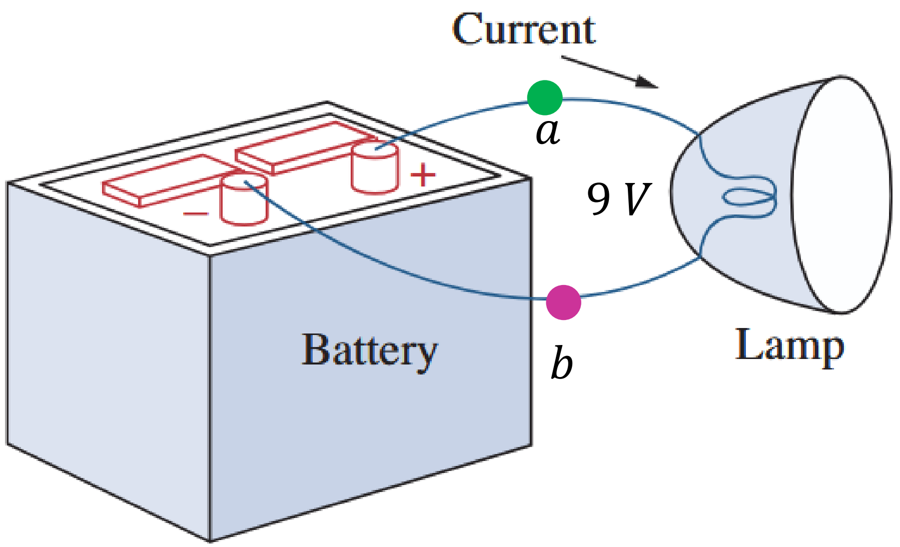

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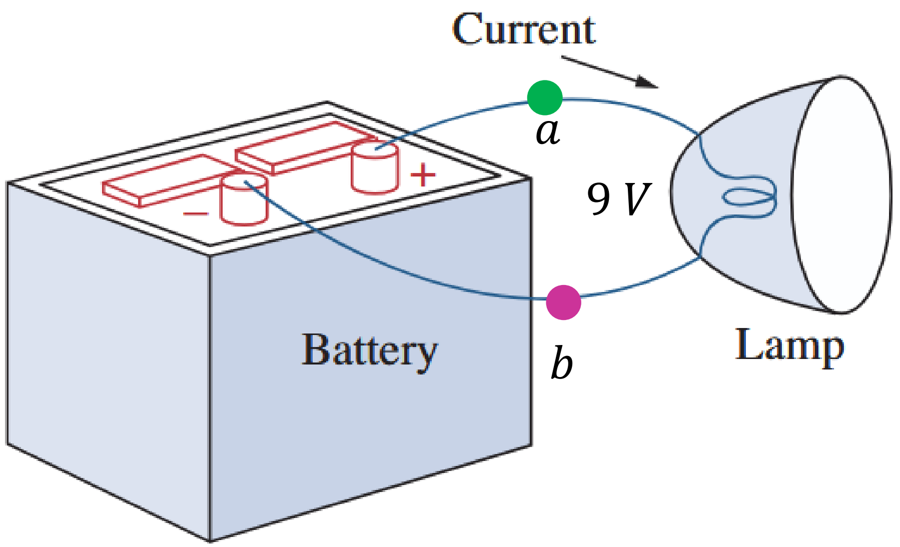

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Voltage is also called "**Potential Difference**"

$$V_a = V_b + 9\ V$$
$$v_{ab} = V_a - V_b = 9\ V$$

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Net rate of flow (of charge)

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A neutral atom flowing would have zero net charge, leading to zero current!

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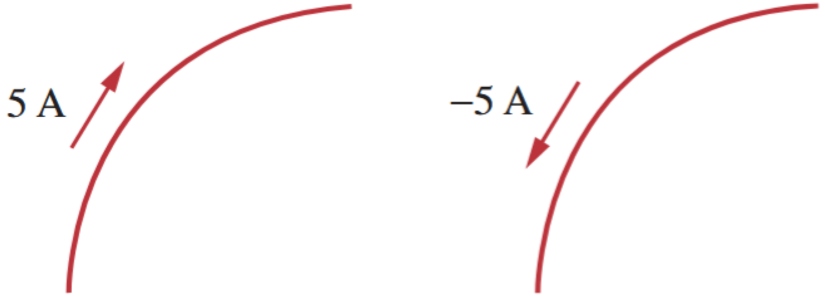
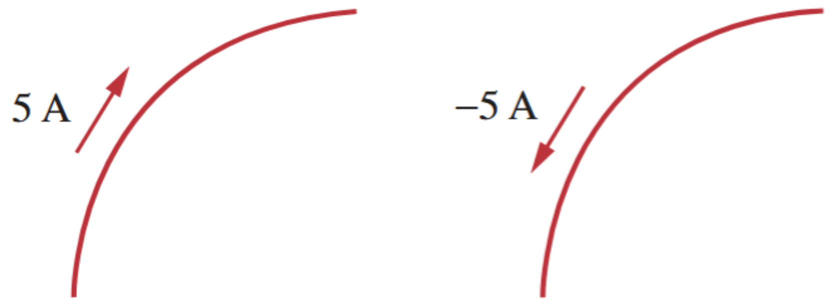


"Current"

Diagrammatically…



We show current by both magnitude and a direction.

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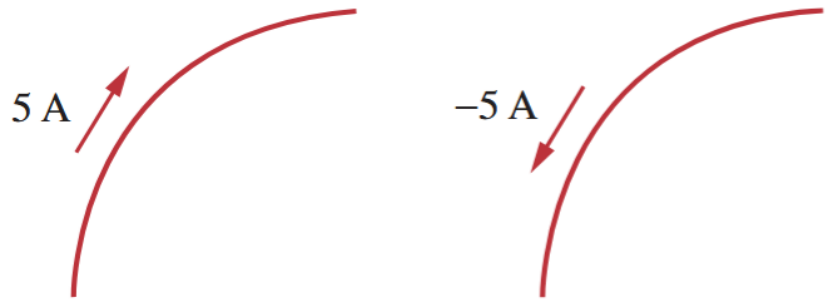
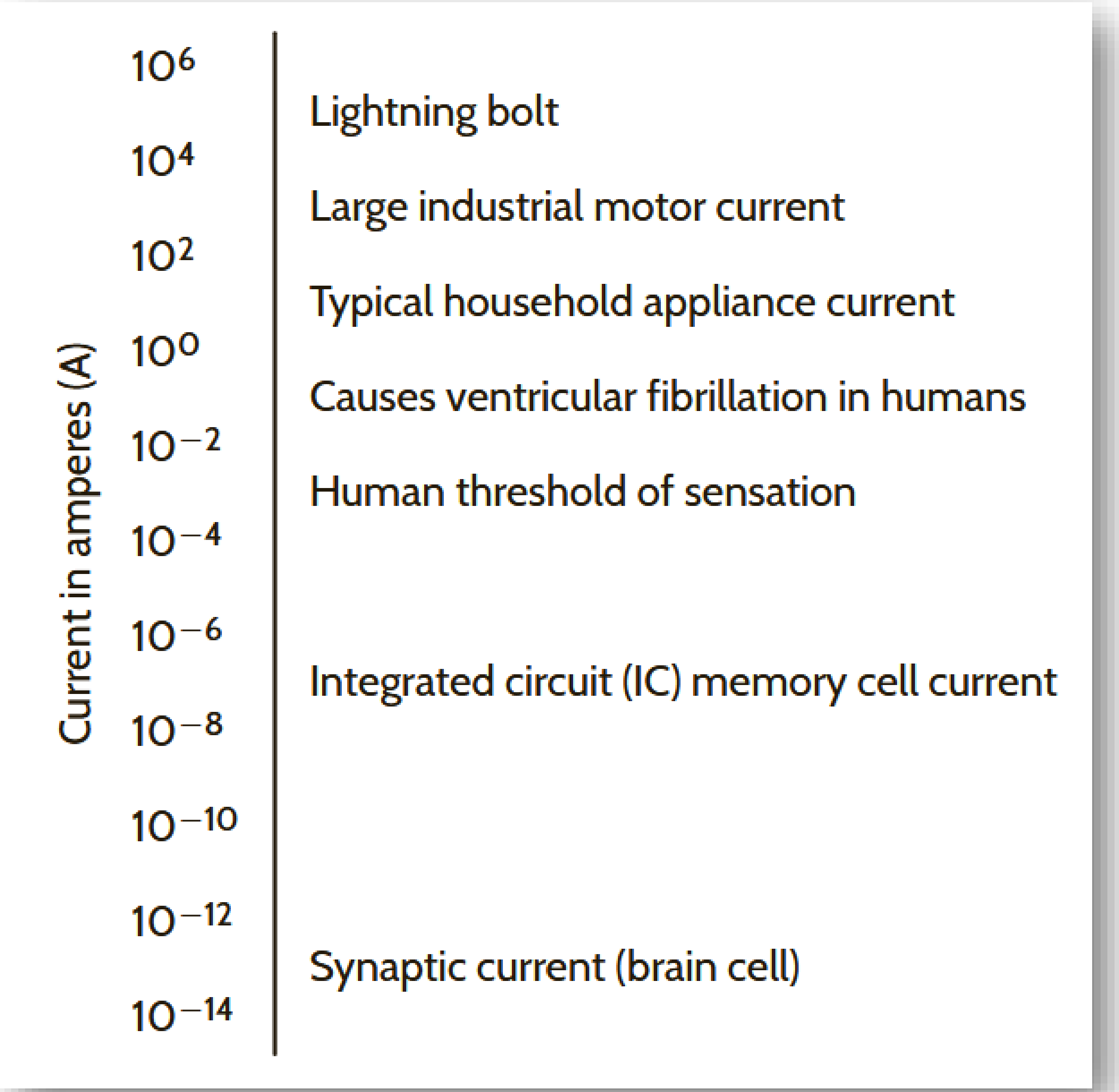


We show current by both magnitude and a direction.

Note that since current is the net flow of charge, a flow of $5\ A$ in one direction is the same as a flow of $-5\ A$ in the opposite direction.

"Current"



Current in amperes (A)
10^6
10^4
10^2
10^0
10^−2
10^−4
10^−6
10^−8
10^−10
10^−12
10^−14
Lightning bolt
Large industrial motor current
Typical household appliance current
Causes ventricular fibrillation in humans
Human threshold of sensation
Integrated circuit (IC) memory cell current
Synaptic current (brain cell)

“Current”

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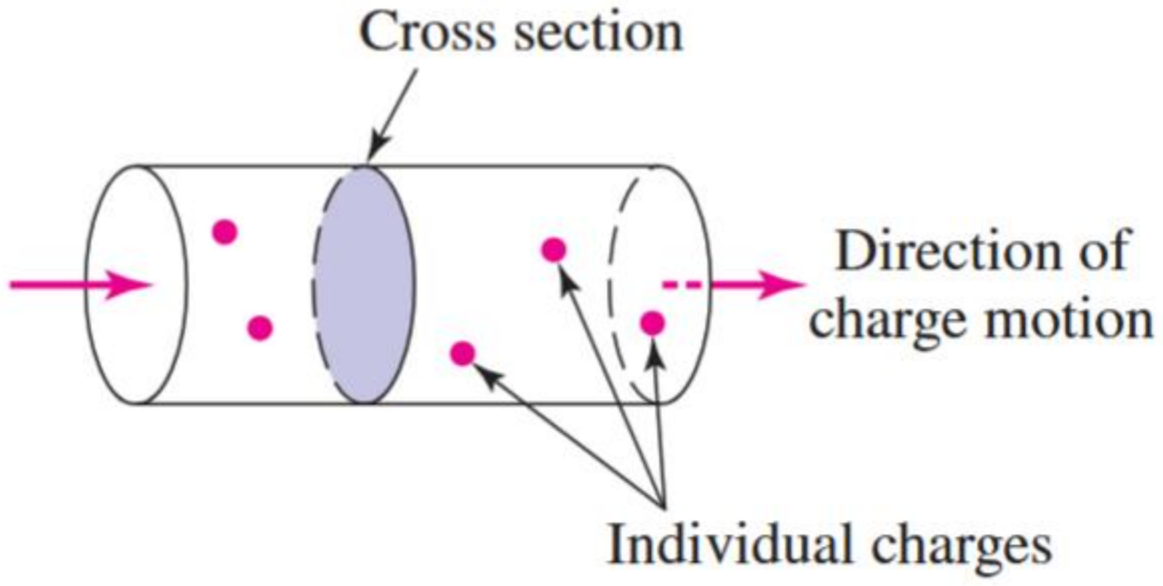

FIGURE 2.1 The definition of current illustrated using current flowing through a wire; 1 ampere corresponds to 1 coulomb of charge passing through the arbitrarily chosen cross section in 1 second.

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- Current Can be Fixed or Time-Varying!

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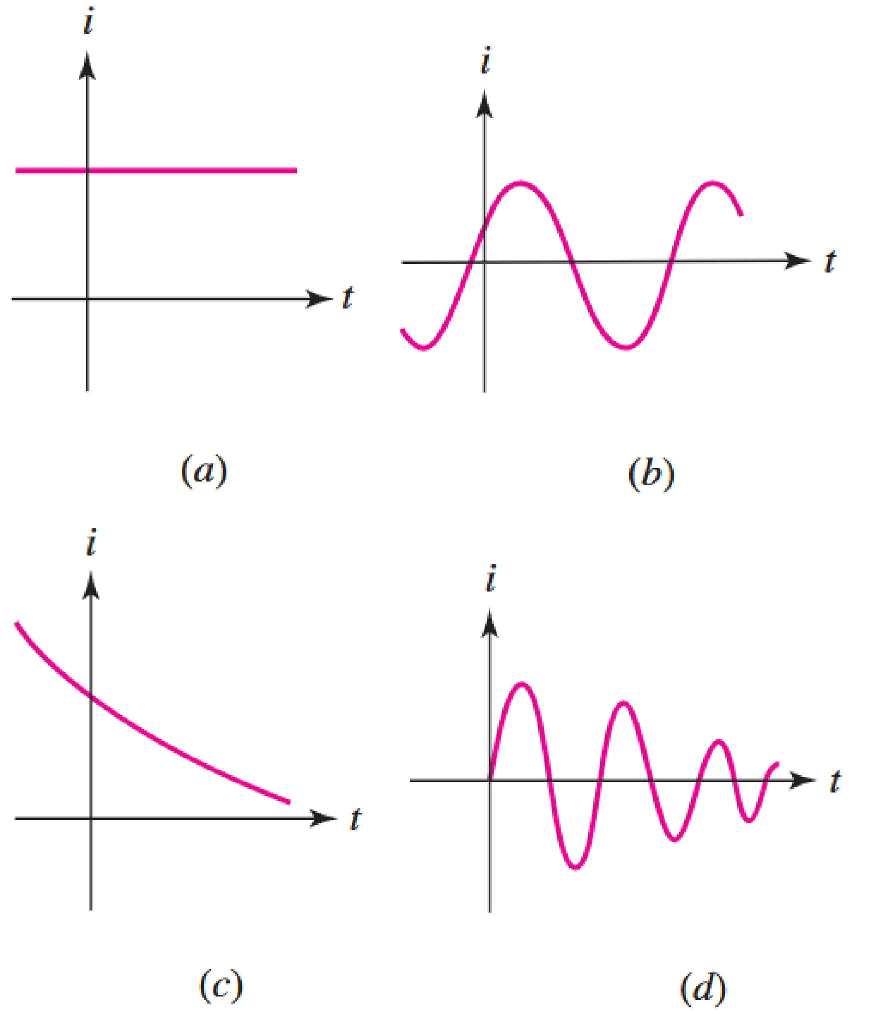
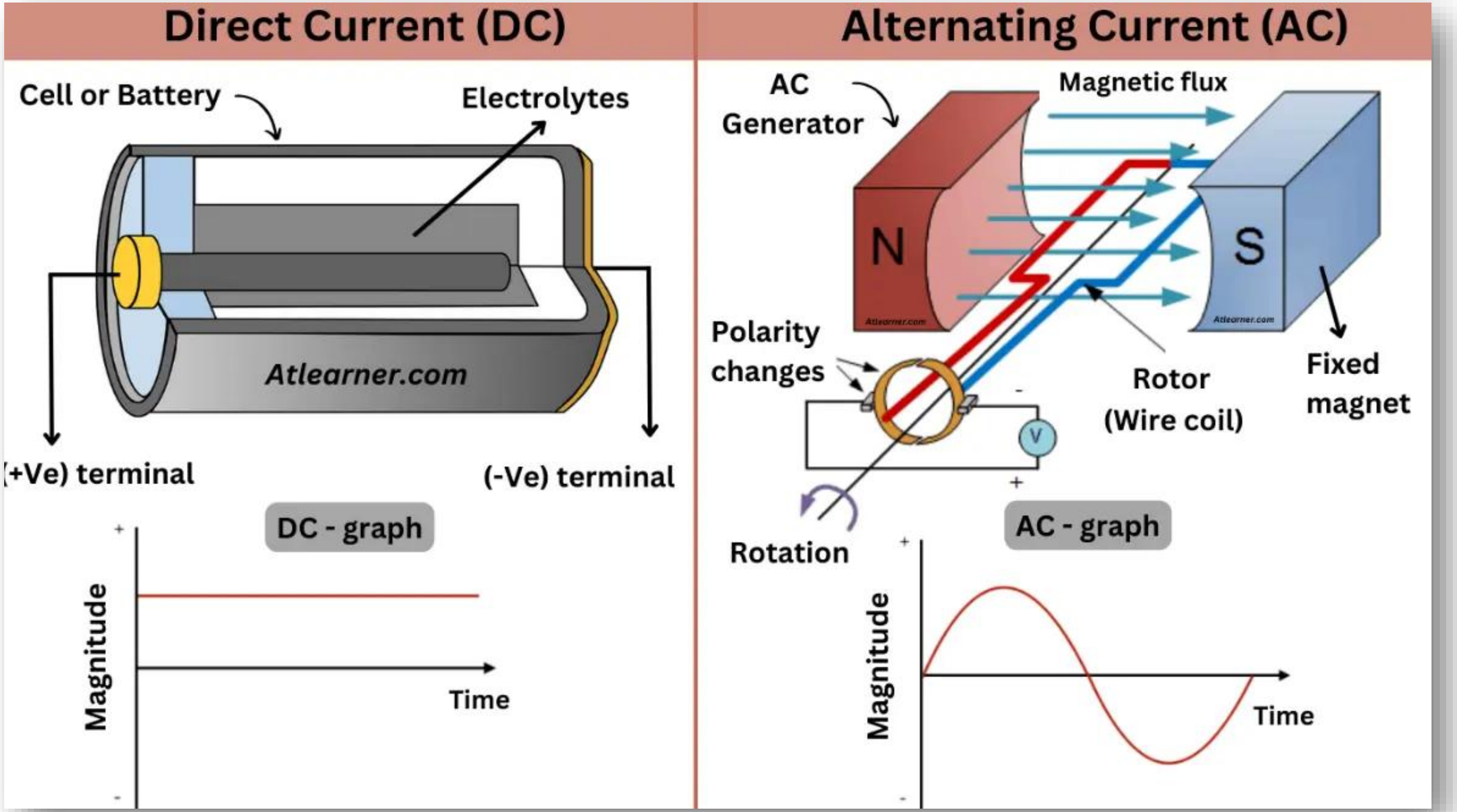


FIGURE 2.4 Several types of current: (*a*) Direct current (dc). (*b*) Sinusoidal current (ac). (*c*) Exponential current. (*d*) Damped sinusoidal current.

The two most common types of current are AC and DC




Electrical appliances
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Electronic and battery-operated devices are typically DC.

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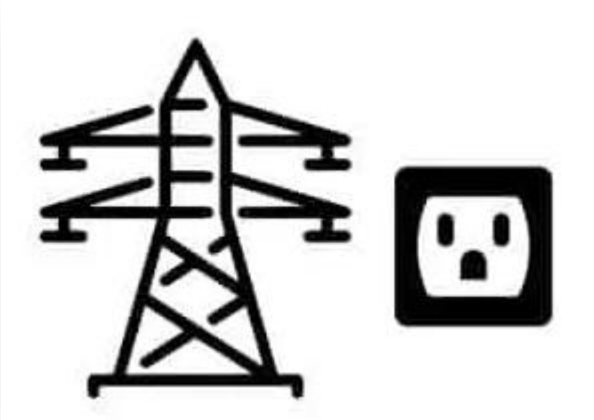


Power transmission systems and your home socket are also AC.

Why? Answers in Electric Circuits-II!

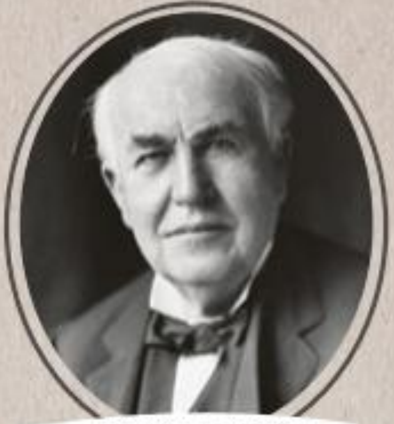
HW: Read up on the **AC-DC War** and how the two may actually complement each other now!

THE GREAT WAR OF THE CURRENTS

AC vs DC

POWER TRANSMISSION

NIKOLA TESLA



THOMAS EDISON

When projecting Europe's future electricity infrastructure systems, discussions revolve around two possible transmission technologies: Alternate Current (AC) and Direct Current (DC).

Today, the two technologies are seen as complementary: AC power transmission is suitable for relatively short connections while DC is the most appropriate technology for carrying high power over long distances practically without any losses. At present, Europe's networks are based on meshed AC electricity grids. In the future, these will be complemented by a high power DC overlay network. While DC connections today are typically used to transmit or exchange power point-to-point, meshed or multi-terminal DC grids will become available as HVDC switchgear technology gains market access.

This complementarity was not always the case. In the late 19th century, Nikola Tesla laid the foundations of today's alternating current electricity supply systems. Around the same time, Thomas Edison set up an extended DC system for the illumination of New York City.

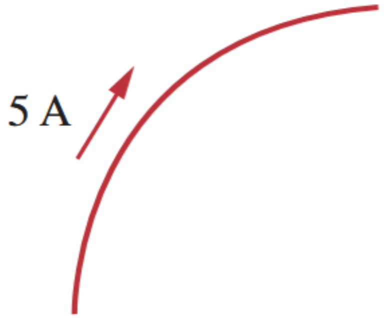
Today Tesla's AC systems operate Europe's electricity systems but we can be confident that Edison's dream to create a DC overlay net will become a reality in Europe in the near future. With that, the fierce "war of the currents" fought between two of the world's most ingenious minds will have been settled in a peaceful co-existence.

We’ve seen two conventions so far, let’s look at a third one…

Conventional Current

What’s Flowing?

- Recall that we represented current by a magnitude and direction of flow.



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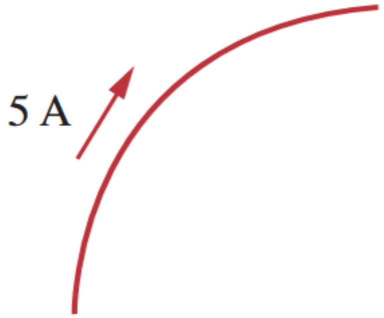

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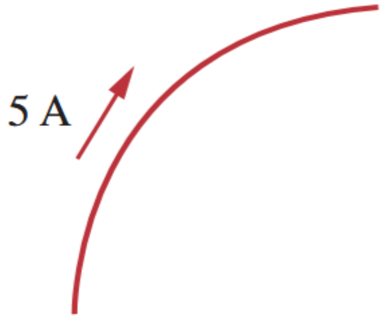

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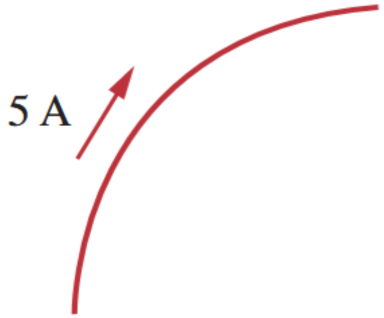
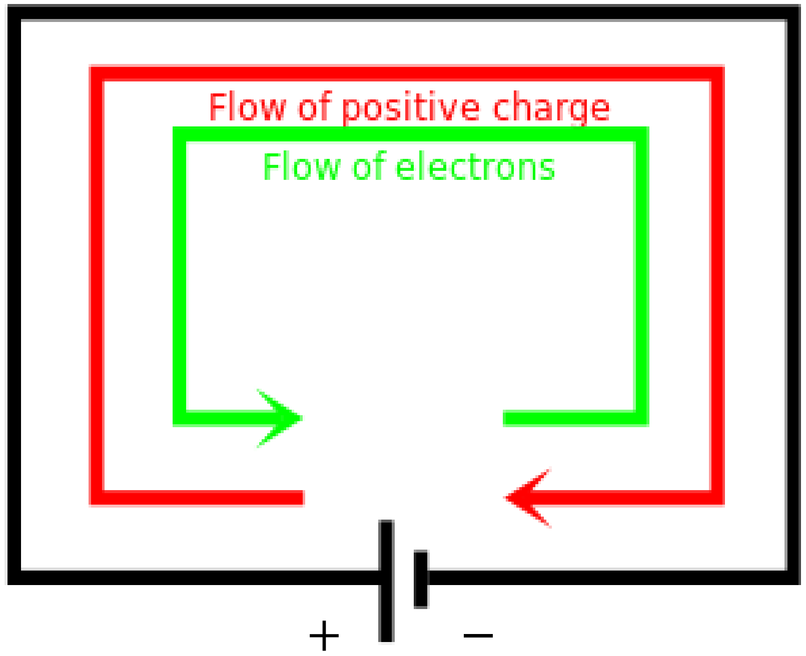
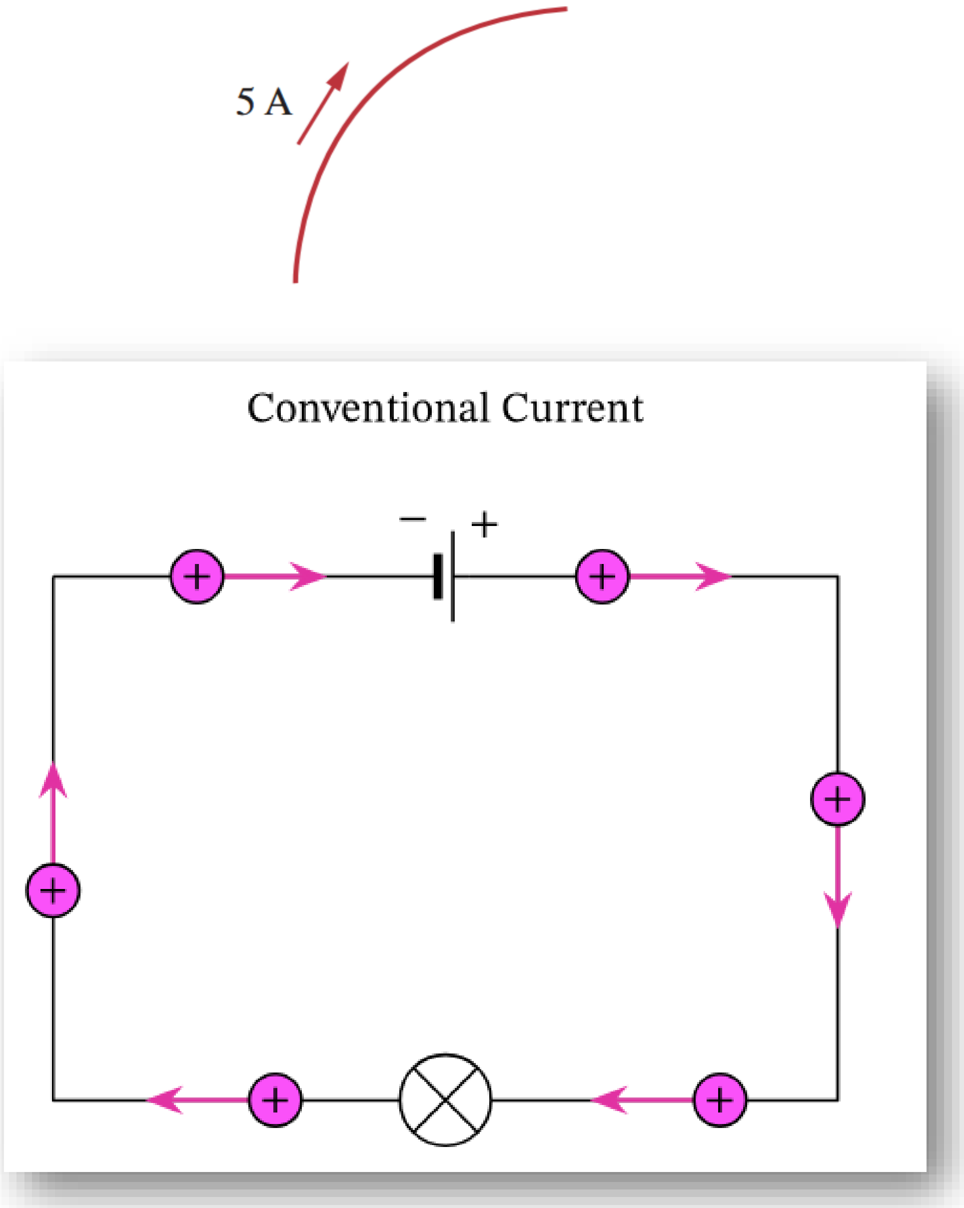
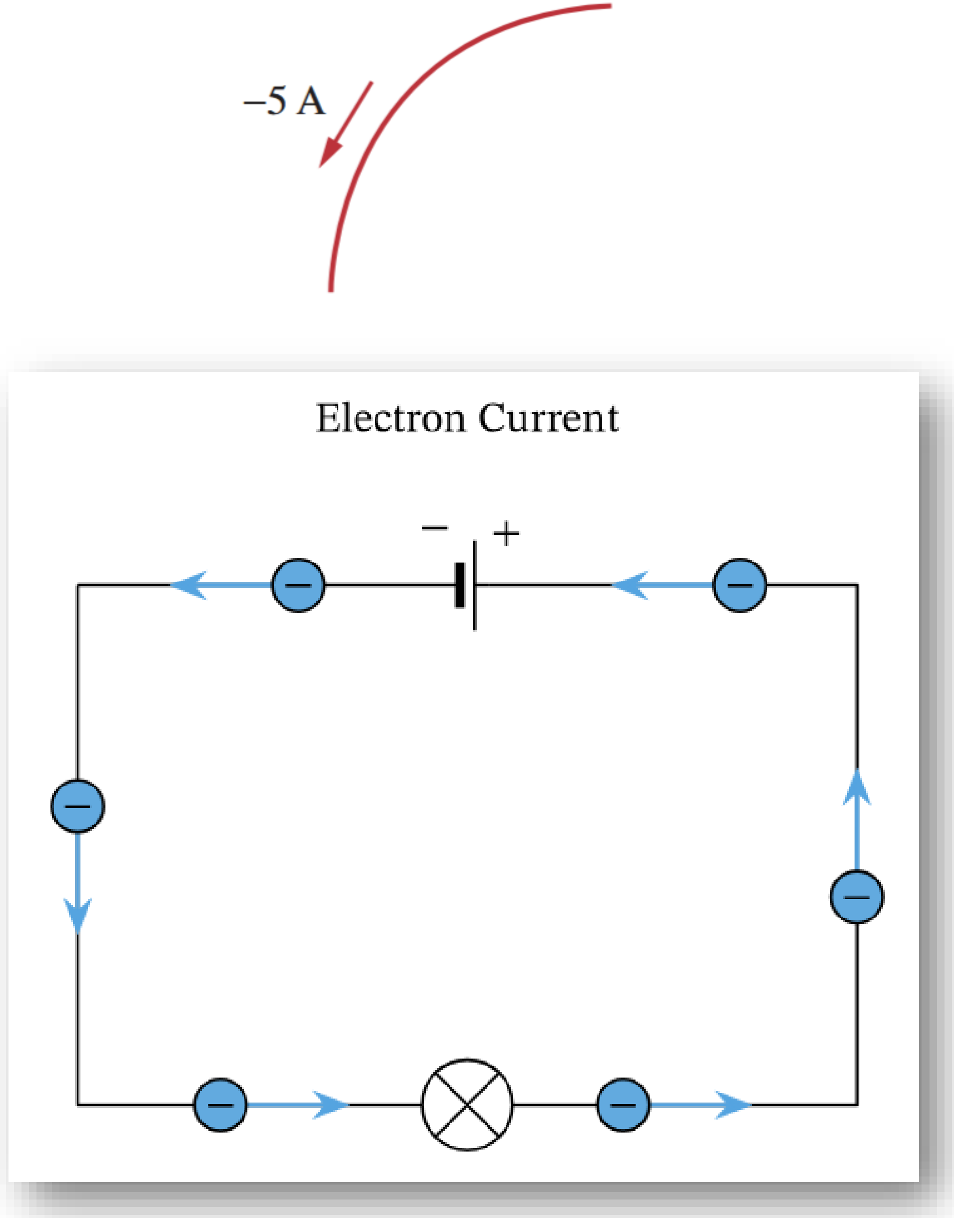
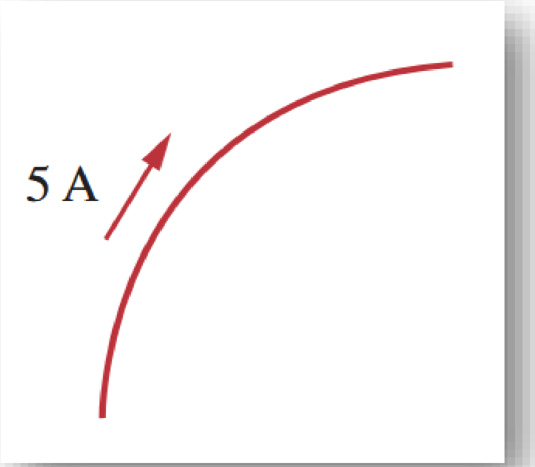
- We can get away with it, as the flow of positive charge in one direction is equivalent to flow of negative charge in the opposite direction!






5 A
Conventional Current
−
+


−5 A
Electron Current
−
+



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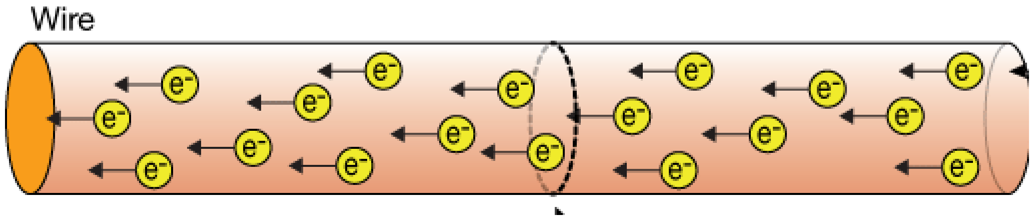

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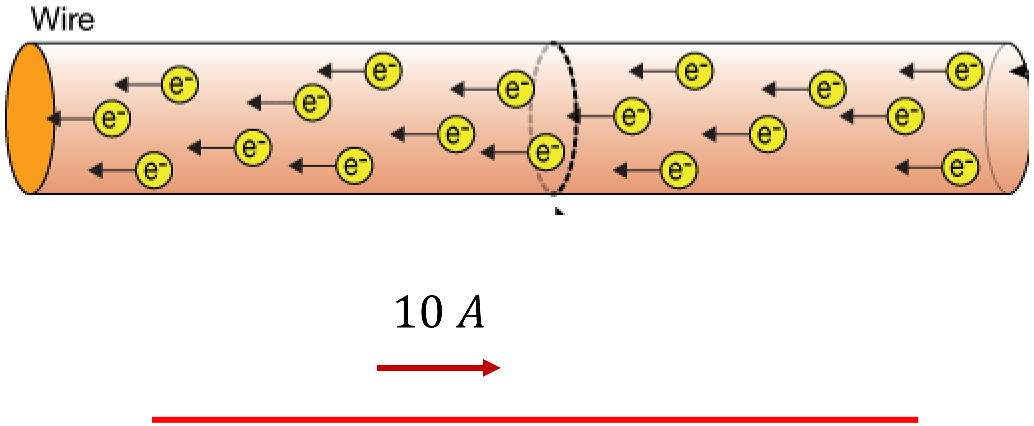
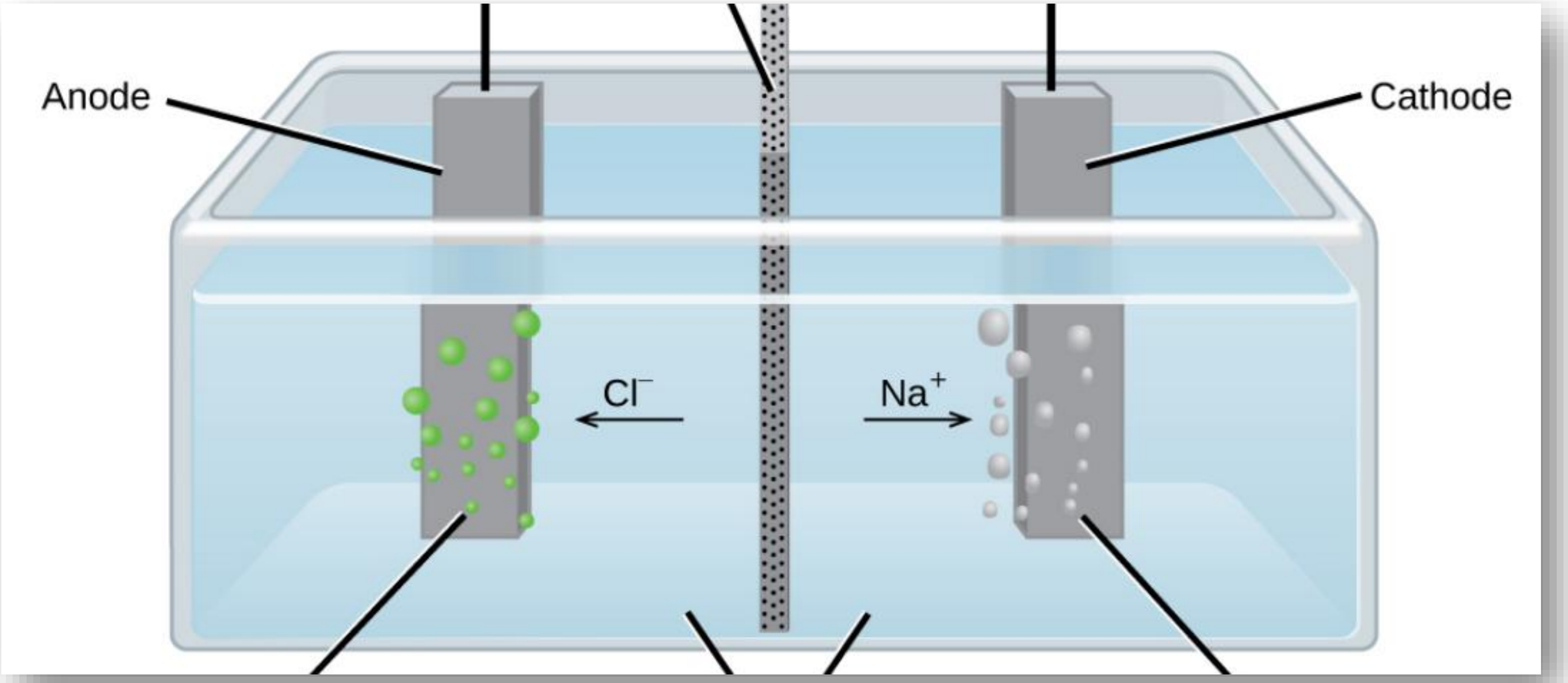


Q. Suppose that $4\ C$ of positive ions are flowing to the right and $8\ C$ of negative ions are flowing to the left per second in this electrolyte. How would you show current flow in the electrolyte for this case?



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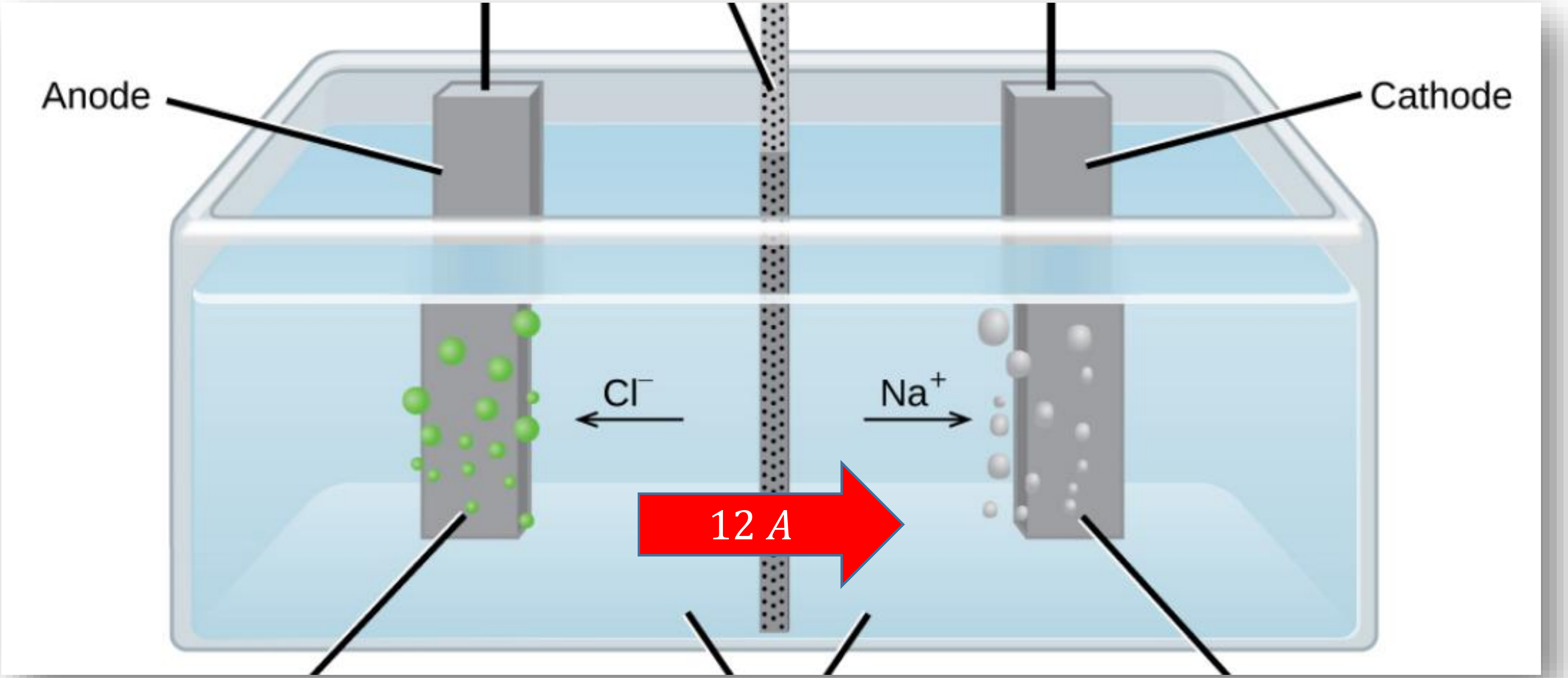

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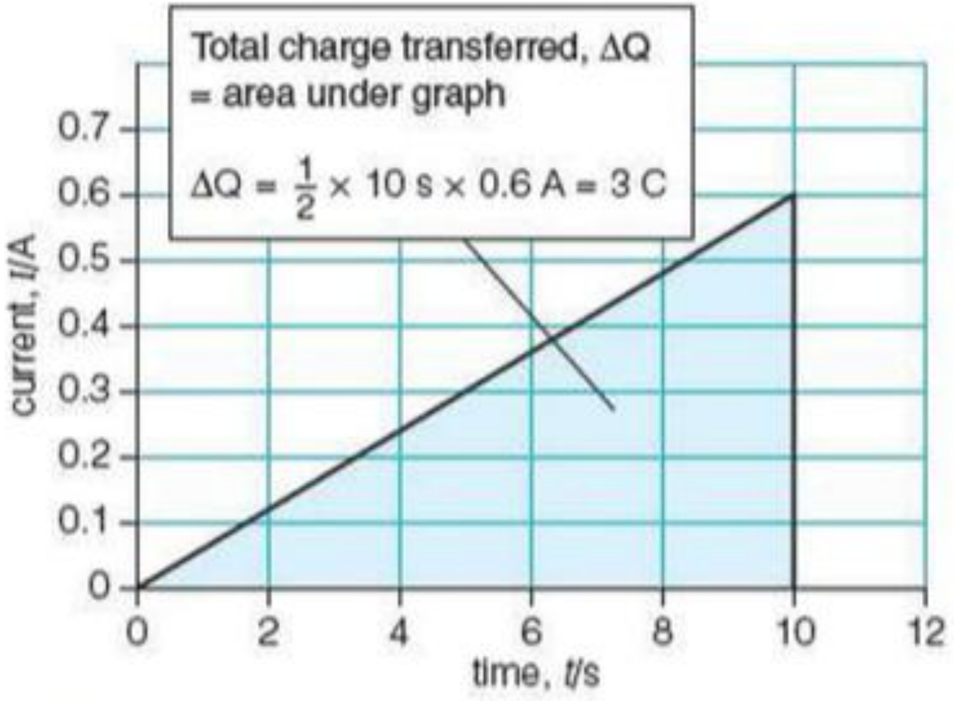

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1 volt = 1 joule/coulomb = 1 newton-meter/coulomb

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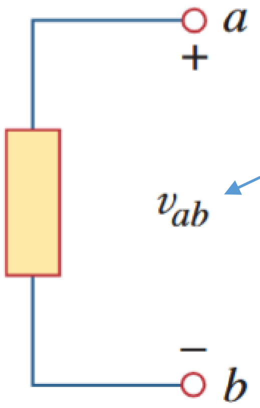

voltage at a w.r.t. b

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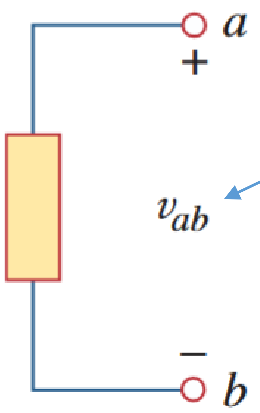

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Keep in mind that

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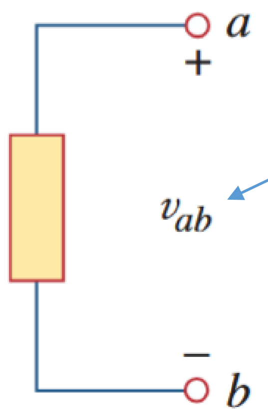

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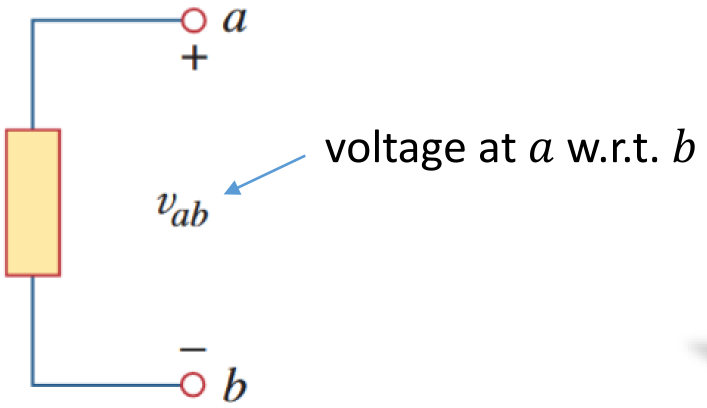

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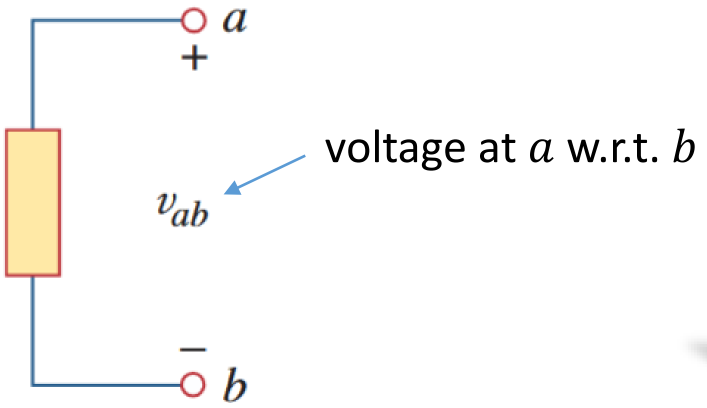

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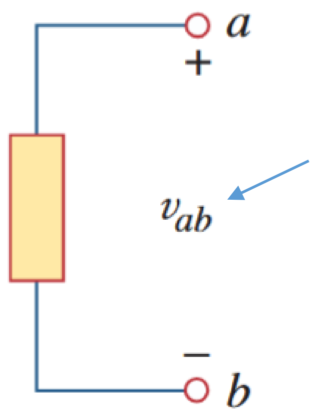

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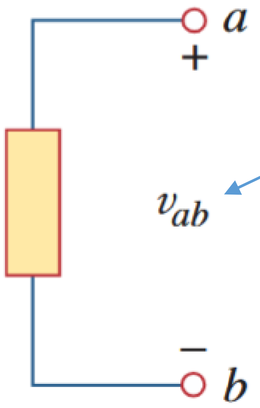

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$$v_{ab} = -v_{ba}$$

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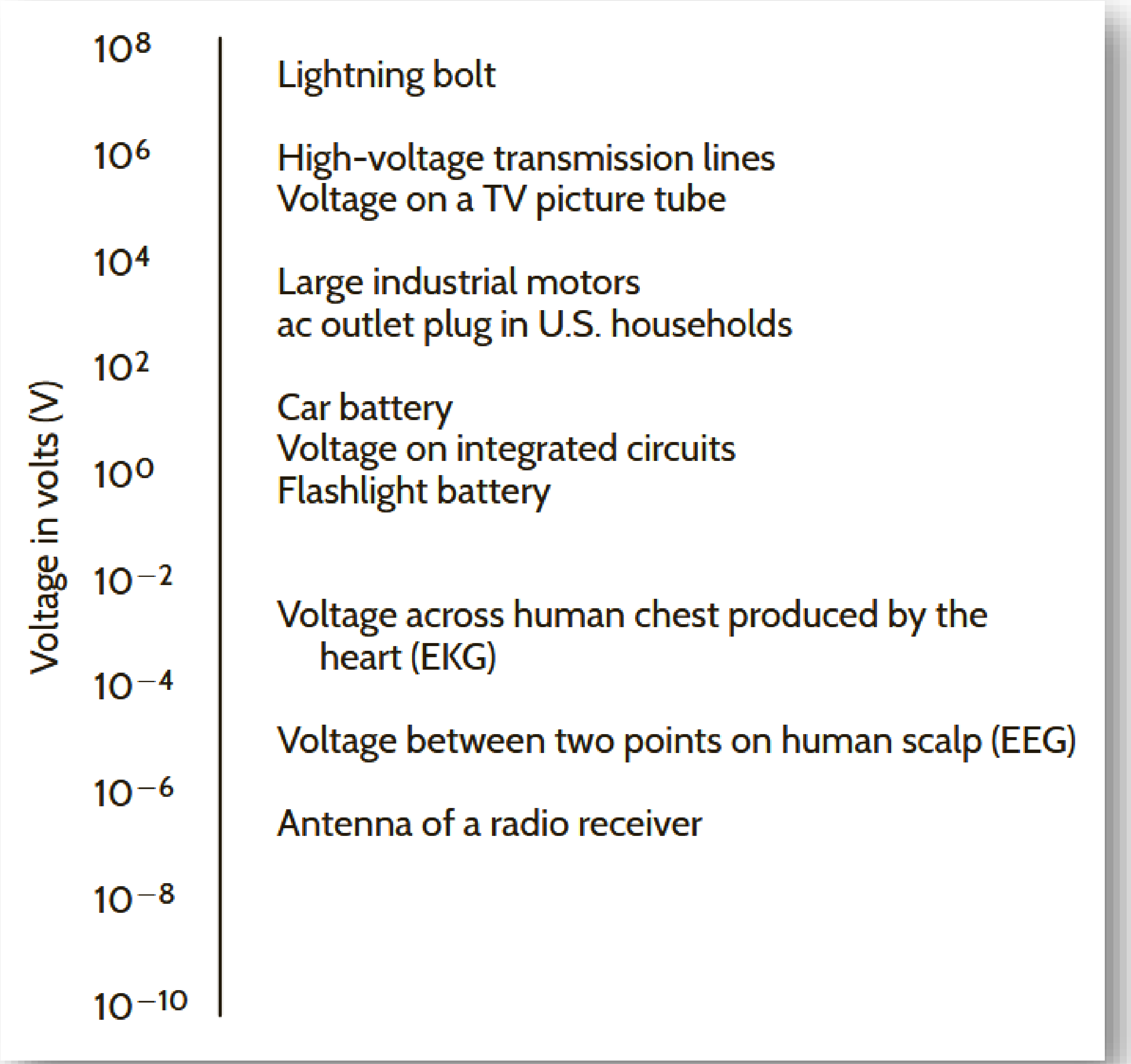
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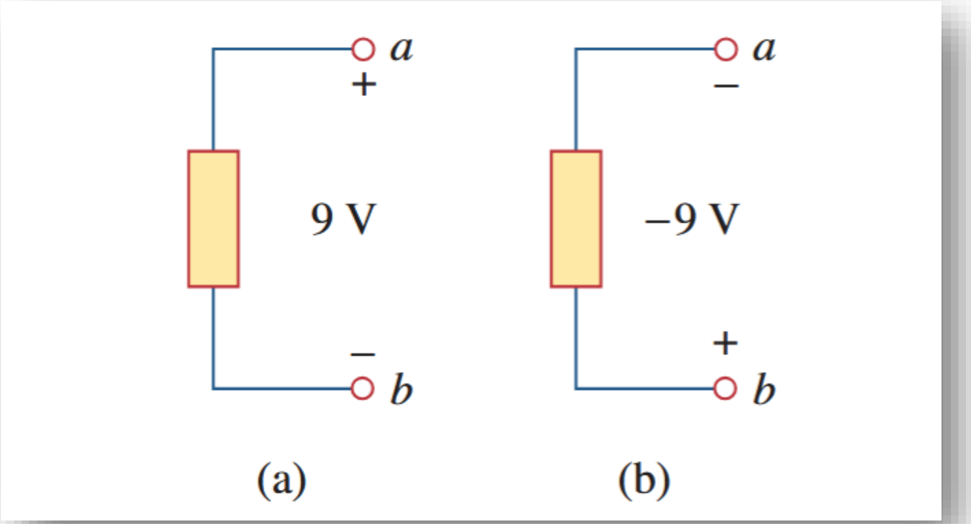
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
Lightning bolt
High-voltage transmission lines
Voltage on a TV picture tube
Large industrial motors
ac outlet plug in U.S. households
Car battery
Voltage on integrated circuits
Flashlight battery
Voltage across human chest produced by the heart (EKG)
Voltage between two points on human scalp (EEG)
Antenna of a radio receiver

A fourth convention…

A truly positive terminal is at a higher potential



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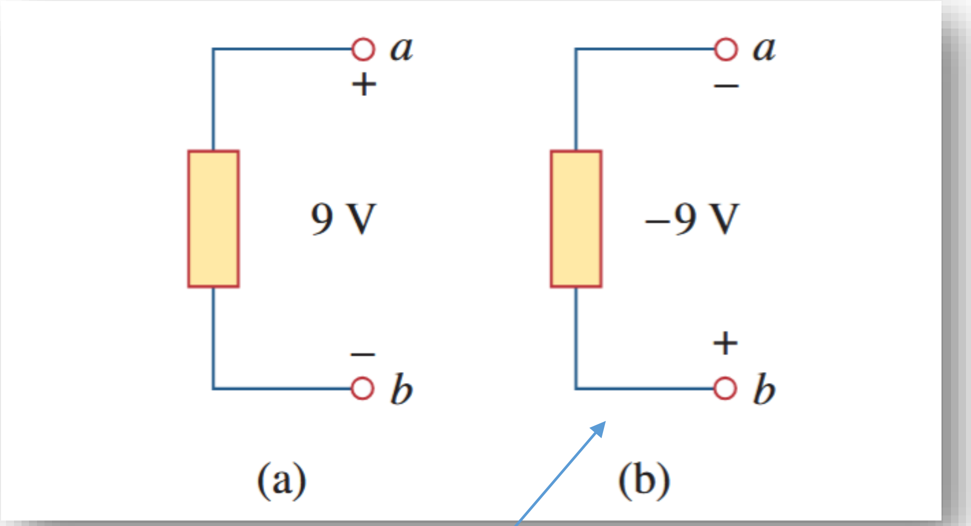

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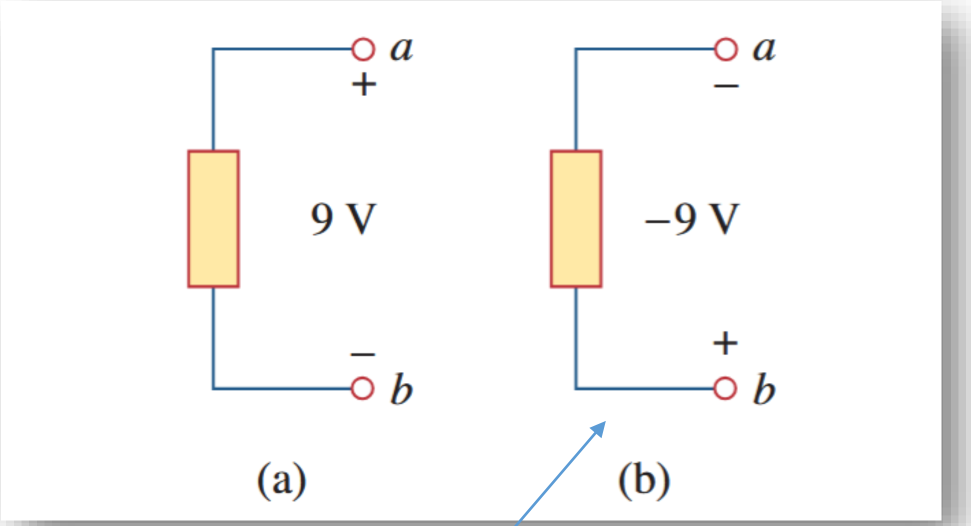
Why?



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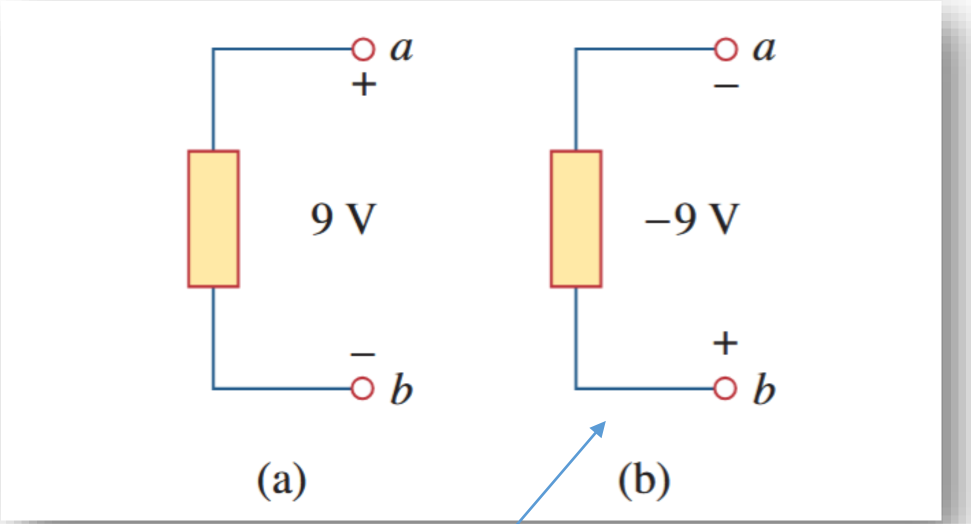

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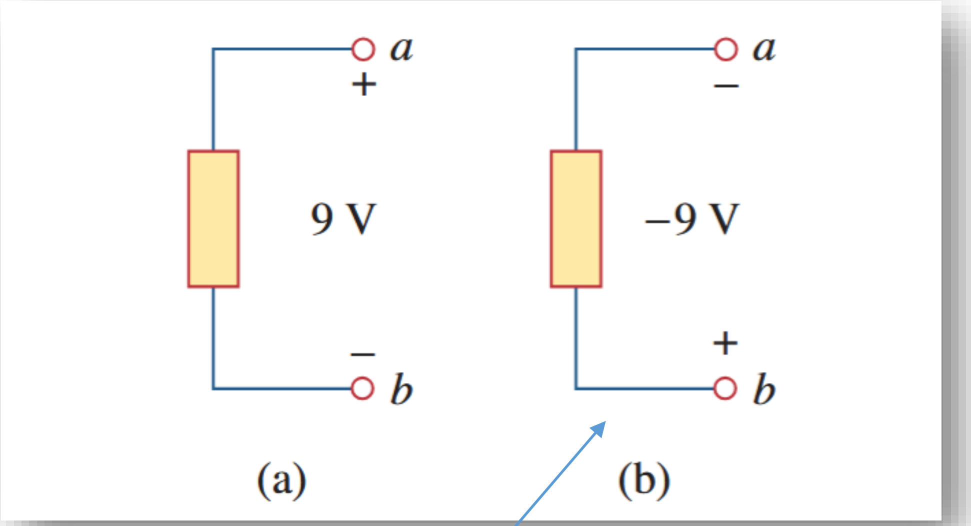

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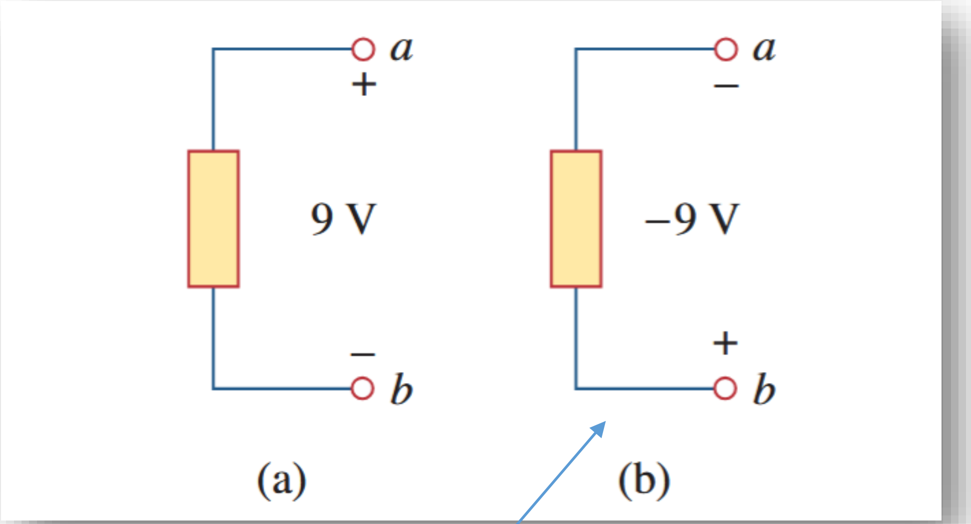

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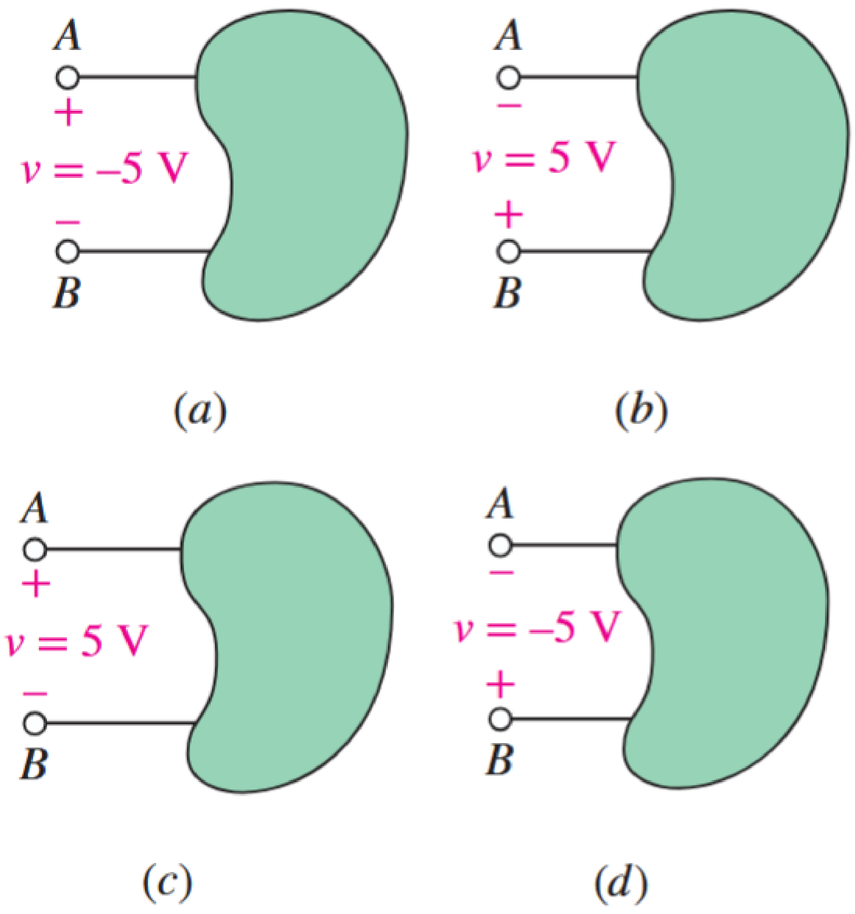
    - i.e., for a positive charge $V_a > V_b$, leading to $v_{ab} = V_a - V_b > 0$
- If we mistakenly assign + to a terminal at low potential, then this will be made clear to us by the voltage turning out negative
    - As then $v_{ba} = V_b - V_a < 0$

Q. Can you describe these in words?



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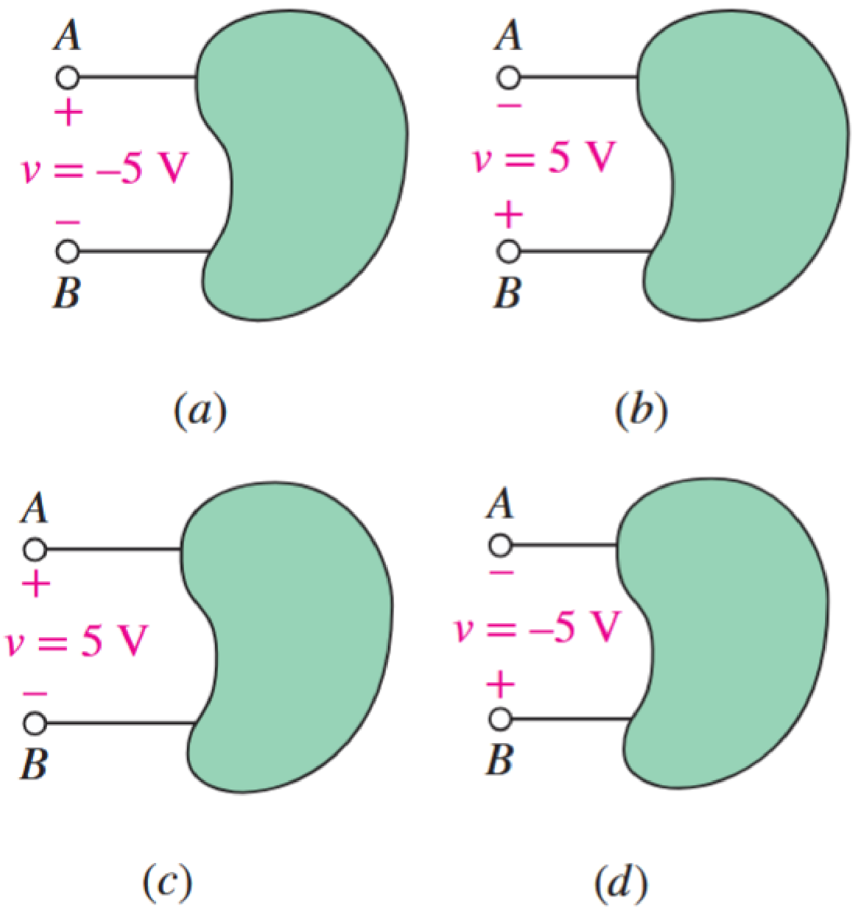

(a) Potential at B is $5\ V$ higher than at A

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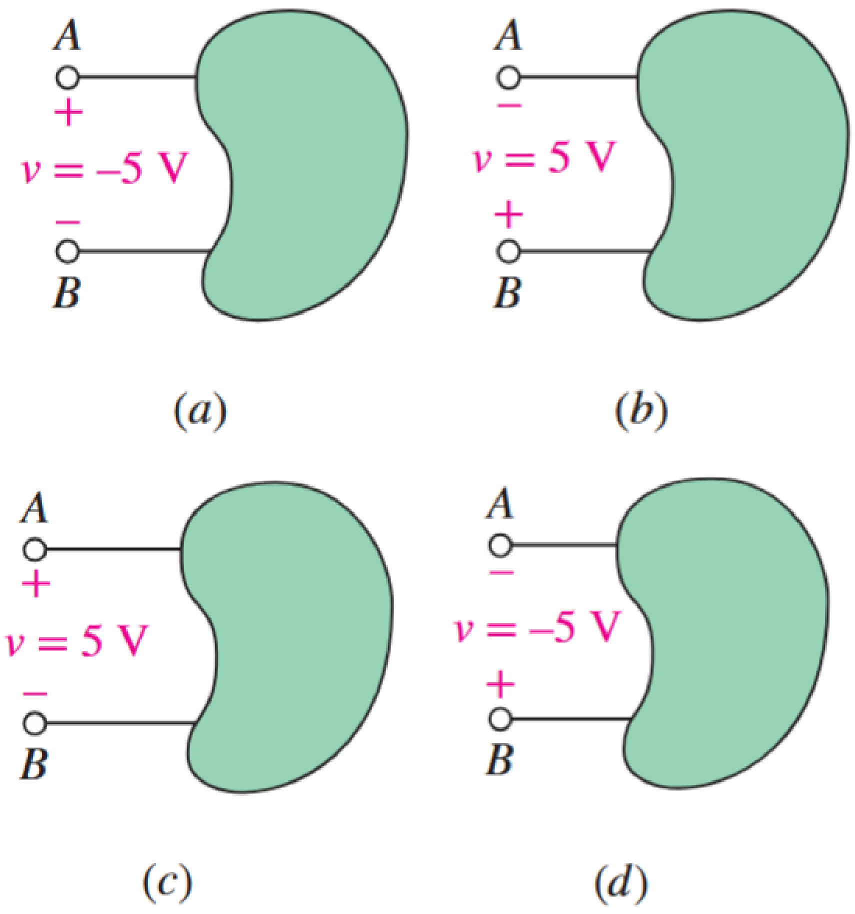

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(b) Potential at B is 5 V higher than at A

(c) Potential at A is 5 V higher than at B

(d) Potential at A is 5 V higher than at B

Next time, we look at more terms and conventions…

Questions?? Thoughts??